The Quantization Cliff: Disproportionate Reasoning Degradation at the W4-to-W3 Boundary

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Abstract

Weight-only quantization at 4 bits (W4A16) generally preserves LLM accuracy, but reducing to W3A16 triggers a *quantization cliff* that disproportionately destroys multi-step reasoning. We systematically document this cliff across seven models spanning five model families (7B–14B) and five benchmarks (GSM8K, MMLU, ARC-Challenge, MATH-500, HumanEval), revealing a three-tier degradation spectrum: two models collapse to 0% GSM8K accuracy with degenerate looping (Llama-3.1-8B at -87 pp, Qwen3-8B at -93.6 pp); three exhibit severe cliffs (-23.8 to -28.2 pp); and two show mild degradation (-3.4 to -11.9 pp). To isolate reasoning-specific damage, we introduce the *Reasoning Excess* (RE) metric—the ratio of GSM8K accuracy drop to MMLU accuracy drop under identical quantization. Three cliff-regime models show RE significantly > 1 after Bonferroni correction ($p < 0.001$, m=5), spanning $2.51\times$ to $4.43\times$; a fourth (Qwen2.5-7B) shows a trend ($p = 0.016$ uncorrected). Collapsed Llama-3.1-8B reaches $5.47\times$ (MMLU at 52.4% despite GSM8K $=$ 0%); only Gemma-2-9B (RE $=$ 0.65) inverts the pattern, losing more knowledge than reasoning. Per-sample flip analysis confirms that longer reasoning chains are significantly more likely to flip from correct to incorrect under quantization in three of five non-collapsed models; partial-correlation analysis across five models disentangles token length from reasoning depth, identifying cumulative token-level error—not discrete chain steps—as the dominant predictor of flip probability. In DeepSeek-R1-7B, MATH-500 difficulty stratification shows cliff severity scaling sharply with problem difficulty (-14 pp at Level 1 to -52 pp at Level 5). Finally, calibrated perturbation replay on three models shows that structured quantization error causes 3–5$\times$ less reasoning degradation than magnitude-matched random noise, establishing that quantization damage depends on error–architecture interaction, not magnitude alone.

1 Introduction

Large language models (LLMs) have achieved remarkable capability on multi-step mathematical reasoning, with competitive 7–14B models reaching 78–94% accuracy on GSM8K [Cobbe et al., 2021] in full precision. Deploying these models at scale requires aggressive compression, and weight-only quantization at 4 bits per parameter (W4A16) has emerged as the practical sweet spot, preserving accuracy within 1–2% across diverse benchmarks [Liu et al., 2025a, Lin et al., 2024, Frantar et al., 2023]. But what happens when we push one bit further?

The answer is not a gentle decline—it is a *cliff*. We evaluate seven models from five families (7B–14B) on GSM8K under W3A16 GPTQ and observe a three-tier degradation spectrum: collapse

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(Llama-3.1-8B -87 pp, Qwen3-8B -93.6 pp to 0%), severe cliff (Qwen2.5-14B -28.2 pp, Phi-3 -26.6 pp, DeepSeek-R1-7B -23.8 pp), and mild (Qwen2.5-7B -11.9 pp, Gemma-2-9B -3.4 pp). This is not proportional degradation—it is a **quantization cliff** where a single bit of additional compression can destroy multi-step reasoning, with severity determined by model family rather than parameter count.

The cliff is *reasoning-specific*. We formalize this as the *Reasoning Excess* (RE) ratio. RE exceeds 1.0 in most non-collapsed models, significantly so in three after multiple-testing correction ($p < 0.001$, Bonferroni $m{=}5$). The sole exception is Gemma-2 (RE $= 0.65$), whose MMLU drop exceeds its mild GSM8K decline—knowledge is more affected than reasoning in this mild-regime model.

This work. We introduce a multi-level diagnostic framework that reveals hidden reasoning failures invisible to perplexity-based evaluation, validated across seven models from five families and five benchmarks (GSM8K, MMLU, ARC-Challenge, MATH-500, HumanEval). Our findings build a layered picture of the cliff:

1. **The cliff exists and is architecture-determined (§4.1).** The W4$\to$W3 transition produces a three-tier spectrum—collapse, cliff, and mild—where severity tracks model family, not scale.
2. **It selectively targets reasoning (§4.2).** Our Reasoning Excess (RE) metric shows reasoning drops 1.26–$5.47\times$ more than knowledge across models; even Llama-3.1-8B's MMLU stays at 52.4% while GSM8K falls to 0%.
3. **Damage accumulates through chains (§4.3).** Per-sample flip rates reach 31%; in 3 of 5 models, longer chains flip disproportionately ($p < 0.01$), consistent with per-step error accumulation.
4. **Token length, not reasoning depth, drives vulnerability (§5.2).** Partial correlations across five models show generation length is the dominant predictor of flip probability after controlling for chain steps; the reverse partial is zero or negative, confirming cumulative token-level error as the mechanism.
5. **Error structure, not magnitude, determines severity (§4.4).** Gaussian noise causes 3.3–$5.6\times$ more damage than structured GPTQ error on Gemma-2; DeepSeek collapses under random noise but not structured noise of equal magnitude. Perturbation replay reproduces the actual GPTQ cliff within 0.2 pp for Gemma-2 and Phi-3; within 4.2 pp for DeepSeek (gap attributable to protocol differences, §4.4).
6. **Mid-layer vulnerability is consistent across architectures (§5).** Layer-level perturbation replay localizes damage to intermediate transformer layers across DeepSeek (-5.7 pp), Phi-3 (-5.3 pp), and Gemma-2 (-4.0 pp), with front layers most resilient in all three models.

Together, these findings provide mechanistic evidence that quantization errors accumulate along reasoning chains, with their impact determined by error structure and concentrated in mid-layer representations where multi-step reasoning is assembled.

2 Related Work

LLM weight-only quantization. Post-training weight quantization compresses LLM parameters to low-bit integers without retraining. GPTQ [Frantar et al., 2023] and AWQ [Lin et al., 2024] establish 4-bit as the production standard; sub-4-bit methods include QuIP# [Tseng et al., 2024], AQLM [Egiazarian et al., 2024], and SqueezeLLM [Kim et al., 2024], while mixed-precision strategies [Dong et al., 2020, Li et al., 2021, Guan et al., 2024] allocate bits non-uniformly via sensitivity metrics (see Gong et al. [2024] for a survey). These methods optimize for perplexity or aggregate accuracy; none examines the sharp reasoning cliff between 4-bit and 3-bit that we characterize.

Quantization effects on reasoning. Liu et al. [2025a] provide the closest prior work, establishing that W4A16 is near-lossless while W3A16 causes significant reasoning degradation across several models. Our work extends their findings along six axes: (i) the RE metric isolating reasoning-specific damage from general compression loss, (ii) per-sample flip analysis with chain-length correlation revealing error accumulation, (iii) perturbation replay decomposing magnitude vs. structure contributions, (iv) coverage of five architecture families including reasoning-distilled DeepSeek-R1, (v) a degenerate output taxonomy distinguishing collapse modes, and (vi) cross-method comparison (GPTQ/RTN/HQQ) disentangling precision from algorithm. Concurrent work examines related phenomena: Li et al. [2025b] document reasoning degradation under low-bit quantization and propose recovery via task-specific fine-tuning; Zhang et al. [2025a] study compression effects on reasoning models including DeepSeek-R1 across quantization, distillation, and pruning; Shi and Ding [2025] provide a broad characterization of quantization methods but without reasoning-specific metrics; Han et al. [2026] identify a "quantization trap" where reduced precision breaks linear scaling laws for multi-hop reasoning, focusing on energy and hardware efficiency trade-offs rather than error mechanisms. Li et al. [2025a] introduce step-level error attribution and in-situ correction for quantization-induced math degradation—a complementary approach that localizes damage within the reasoning chain, whereas our perturbation replay localizes damage across architectural modules. Zhou et al. [2026] decompose quantization damage into *signal degradation* (gradual accuracy loss) and *computation collapse* (catastrophic output failure), identifying these as two distinct failure modes. Our work differs in three ways: (i) we isolate the role of error *structure* vs. magnitude via controlled perturbation replay rather than post-hoc decomposition; (ii) we localize vulnerability to specific layer groups (mid-layer concentration) rather than characterizing global failure modes; and (iii) our RE metric and flip analysis quantify reasoning-*specific* damage relative to knowledge tasks, whereas their taxonomy applies to general generation quality. Federici et al. [2026] provide a theoretical lens on quantization error through signal-to-quantization-noise ratio decomposition into concentration and alignment components; their block transforms target 4-bit inference accuracy, complementing our empirical finding that error structure (not just magnitude) governs reasoning damage at 3-bit. On the KV-cache side, KVSink [Su and Yuan, 2025] and related work [Liu et al., 2025b, Zhang et al., 2025c, Son et al., 2024] develop mixed-precision strategies; our work differs in studying *weight* quantization.

Reasoning evaluation and chain-level analysis. Multi-step reasoning is evaluated on GSM8K [Cobbe et al., 2021] and MATH [Hendrycks et al., 2021] with chain-of-thought prompting [Wei et al., 2022], but aggregate accuracy obscures question-level dynamics. We introduce paired per-question flip analysis into quantization evaluation, separating consistent reasoning loss from noise-induced reshuffling.

Attention sinks and quantization. Attention sinks—tokens absorbing disproportionate attention mass—have been characterized as structural primitives [Gu et al., 2025, Zhang et al., 2025b, Queipo-de Llano et al., 2026]. Our work intersects at the weight level: aggressive weight quantization may perturb the attention distributions that maintain sinks, providing a channel through which quantization disrupts long-range coherence.

Sub-4-bit quantization and QAT. Ouyang et al. [2025] show training maturity modulates quantization sensitivity; QuaRot [Ashkboos et al., 2024] and ParoQuant [Liang et al., 2026] target low-bit inference via rotation; QAT methods [Liu et al., 2023, Shao et al., 2023] may better preserve reasoning but are impractical at 7B+ scale. Recent reasoning-aware approaches include Lv et al. [2026], who systematically study QAT for reasoning LLMs and show that knowledge distillation plus PTQ initialization recovers significant accuracy at 2-bit, and Zhang et al. [2026], who leverage fine-tuning weight-update signals to guide channel-wise quantization for reasoning models. Our cliff characterization provides an evaluation framework for these methods on reasoning tasks beyond perplexity, combining RE to isolate capability-specific damage, flip rate to quantify instability, and perturbation replay to decompose the damage mechanism.

3 Methodology

Our methodology operationalizes four contributions: (C1) characterizing the W4→W3 cliff across 7 models, (C2) isolating reasoning-specific damage via the RE ratio, (C3) localizing damage via flip analysis and chain-length correlations, and (C4) testing architecture-dependence via perturbation replay.

3.1 Model Selection

We evaluate 7 open-weight models (7B–14B) spanning 5 architecture families: Llama-3.1-8B, Qwen2.5-{7B, 14B}, Qwen3-8B, Phi-3-medium-14B, Gemma-2-9B, and DeepSeek-R1-Distill-Qwen-7B (reasoning-distilled from an R1 teacher; full model details in Appendix A.3).

3.2 Quantization Methods

We compare FP16 (gold standard), W4A16 via AWQ [Lin et al., 2024] (4-bit, group size 128, activation-aware scaling, Pile calibration), and W3A16 via GPTQ [Frantar et al., 2023] (3-bit, group size 128, layer-wise optimal brain surgeon, wikitext-2 calibration). The method asymmetry reflects a deployment constraint: AWQ does not support sub-4-bit. GPTQ W4A16 controls on DeepSeek (74.6% vs. AWQ 75.4%, $\Delta = 0.8$ pp) and Qwen3 (92.2% vs. AWQ 92.4%, $\Delta = 0.2$ pp) confirm the cliff is a precision threshold, not a method artifact (Appendix A.16). Our W3A16 results characterize GPTQ-specific behavior; scope caveats in Appendix A.23.

3.3 Evaluation Framework

We evaluate on five benchmarks: **GSM8K** (primary, greedy, `max_new_tokens`=4096), **MMLU** (knowledge control, 5-shot log-likelihood—immune to generation truncation), **ARC-Challenge** (25-shot log-likelihood), **MATH-500** (greedy, 8192 tokens; five difficulty levels test depth-dependent damage), and **HumanEval** (pass@1 on 164 code problems). GSM8K main comparisons (Table 1) use 0-shot evaluation with model-native chat templates: without in-context exemplars, models must rely entirely on internalized reasoning, better exposing quantization-induced degradation. The cross-method analysis (§5.4) uses 8-shot chain-of-thought to match the protocol under which RTN and HQQ baselines were evaluated. Pairing GSM8K with MMLU yields the RE ratio $\mathrm{RE} = \Delta\mathrm{GSM8K}/\Delta\mathrm{MMLU}$. For DeepSeek-R1-7B, we use a 450-question matched subset for accuracy comparisons; behavioral analyses (Appendix A.1) use the full evaluation matched by `question_idx`.

3.4 Flip Analysis

We define the flip rate as the fraction of FP16-correct questions that become incorrect under W3A16:

$$\mathrm{FlipRate} = \frac{|\{i : y_i^{\mathrm{FP16}} = 1,\ y_i^{\mathrm{W3}} = 0\}|}{|\{i : y_i^{\mathrm{FP16}} = 1\}|}. \tag{1}$$

McNemar's test assesses whether correct→incorrect and incorrect→correct flip counts differ significantly. To test depth-dependent damage, we median-split FP16-correct questions by generation length and compare flip rates via two-proportion z-test.

3.5 Perturbation Replay Design

For each linear layer ℓ, we extract the quantization error and inject it at controlled scale:

$$\Delta W^{(\ell)} = W^{(\ell)}_{\text{dequant(W3)}} - W^{(\ell)}_{\text{FP16}}. \tag{2}$$

We inject $c \cdot \Delta W^{(\ell)}$ into FP16 weights via forward hooks ($c \in \{0.25, 0.5, 1.0, 2.0\}$), testing four conditions: baseline (FP16), all_weights (all linear layers), mlp_only, and attention_only. The protocol is applied to three models spanning the severity spectrum: cliff-regime DeepSeek and Phi-3, and mild-regime Gemma-2.

3.6 Profile Extraction

For each model–regime combination, we extract 4D functional profiles (sink concentration, entropy, Gini coefficient, positional bias) per KV-head group, using a sink-token window of k=4. Statistical methods are detailed in Appendix A.22.

Table 1: GSM8K accuracy (%) across seven models and three precision tiers. Δ_{cliff}: FP16→W3A16 drop. All models evaluated on N=500 except DeepSeek (N=450, common subset). The transition splits models into collapse (0%), cliff (−24 to −28 pp), and mild (−3 to −12 pp).

Model	FP16	W4A16 AWQ	W3A16 GPTQ	Δ_{cliff}	Regime
Llama-3.1-8B	87.0	85.6	0.0	−87.0	Collapse
Qwen3-8B	93.6	92.4	0.0	−93.6	Collapse
Qwen2.5-14B	93.6	92.6	65.4	−28.2	Cliff
Phi-3-medium-14B	91.4	—‡	64.8	−26.6	Cliff
DeepSeek-R1-7B	78.0	75.4	54.2⋆	−23.8	Cliff
Qwen2.5-7B	90.9	88.9	79.0	−11.9	Mild
Gemma-2-9B	83.6	83.2	80.2	−3.4	Mild

⋆GPTQ is the worst 3-bit method for this model; RTN (62.6%) and HQQ (84.2%) both outperform it (§5.4).
‡Phi-3 W4A16 omitted: AWQ produces anomalous results; AutoGPTQ and GPTQModel do not reliably support this architecture at W4A16.

4 Results

All GSM8K evaluations use greedy decoding with `max_new_tokens`=4096 (8192 for DeepSeek-R1 structured perturbation replay and MATH-500); GPTQ uses group size 128.

4.1 The Quantization Cliff

Figure 1a and Table 1 reveal the central finding: a single bit of additional compression—from W4A16 to W3A16—triggers a discontinuous transition in reasoning accuracy. W4A16 AWQ preserves near-FP16 performance across all six models tested (within 2.6 pp), but W3A16 GPTQ splits models into three qualitatively distinct regimes: *collapse* (Llama-3.1-8B and Qwen3-8B fall to 0%, producing degenerate loops), *cliff* (Qwen2.5-14B -28.2 pp, Phi-3 -26.6 pp, DeepSeek-R1-7B -23.8 pp), and *mild* (Qwen2.5-7B -11.9 pp, Gemma-2 -3.4 pp). Within the mild tier, severity ranges from near-lossless (Gemma-2, -3.4 pp) to moderate (Qwen2.5-7B, -11.9 pp).

What determines which regime a model falls into? Not scale: Qwen3-8B (tied for the highest FP16 accuracy at 93.6%) collapses completely, while Gemma-2 is least affected. Not quantization method alone: GPTQ W4A16 controls on DeepSeek (74.6%) and Qwen3 (92.2%) match AWQ, confirming a precision threshold (Appendix A.16), though at W3A16 the choice of method matters—RTN (62.6%) and HQQ (84.2%) both outperform GPTQ (54.2%) on the same model (§5.4). Rather, severity tracks architecture family—within the Qwen2 family, all three variants land in the non-collapse regime regardless of scale. The cliff replicates on MATH-500 and HumanEval (Appendix A.24); Gemma-2's MATH-500 cliff (-11.2 pp) is $3.3\times$ its GSM8K cliff, showing that "mild" is task-dependent. HumanEval confirms this pattern ($\Delta = -19.5$ pp, $5.7\times$ the GSM8K cliff; Appendix A.24). Perplexity is an unreliable proxy for reasoning quality: DeepSeek's perplexity rises only 33.8% despite a 23.8 pp accuracy collapse (Appendix A.25).

This cliff is not uniform capability loss. If it were, all benchmarks would degrade proportionally—but they do not.

4.2 The Cliff Targets Reasoning

To measure whether the cliff reflects general compression damage or something specific to reasoning, we define **Reasoning Excess (RE)** as $\Delta_{\text{GSM8K}}/\Delta_{\text{MMLU}}$ under W3A16. RE $= 1$ means reasoning and knowledge degrade equally; RE > 1 means reasoning is disproportionately damaged.

The result is striking (Figure 1b, Table 2). Three cliff-regime models show RE significantly above 1 after Bonferroni correction ($p < 0.001$, m=5); mild-regime Qwen2.5-7B reaches $1.26\times$ ($p = 0.016$ uncorrected, 99% CI contains 1). The most extreme case is collapsed Llama-3.1-8B, where MMLU falls only 15.9 pp (to 52.4%, well above chance) while GSM8K is completely destroyed—RE $= 5.47\times$. This model can still answer knowledge questions but cannot reason through multi-step problems, a dissociation that log-likelihood-scored MMLU (immune to generation degeneration) makes unambiguous. Qwen3-8B presents a contrasting collapse mode: MMLU drops 52.0 pp to 22.9% (near chance), yielding RE $= 1.80\times$—both reasoning and knowledge are

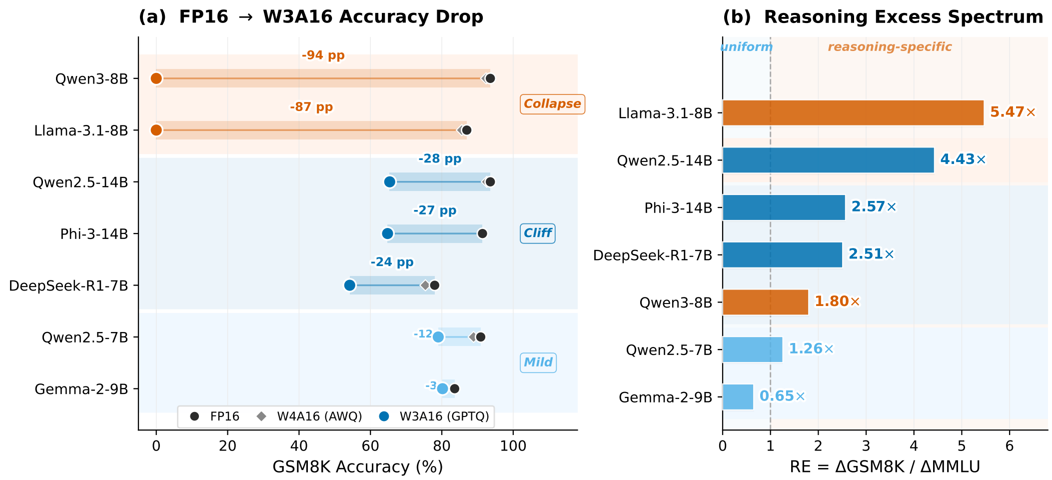

Figure 1: **The Quantization Cliff. (a)** FP16→W3A16 accuracy drop on GSM8K for seven models, grouped by regime. Bar length is proportional to cliff magnitude: collapse models lose 87–94 pp (to 0%), cliff models lose 24–28 pp, mild lose 3–12 pp. Gray diamonds show W4A16 (near-lossless). **(b)** Reasoning Excess spectrum: RE $= \Delta\text{GSM8K}/\Delta\text{MMLU} > 1$ for six of seven models (only Gemma-2 inverts), confirming reasoning-specific damage; Llama-3.1-8B reaches $5.47\times$.

Table 2: Reasoning Excess (RE) under W3A16 GPTQ. Three of five non-collapsed models show RE significantly > 1 after Bonferroni correction ($\alpha = 0.01$, m=5). The two collapsed models diverge: Llama-3.1-8B (RE $= 5.47\times$) retains knowledge while losing reasoning; Qwen3-8B (RE $= 1.80\times$) collapses nearly uniformly. Uniform loss would predict RE $\approx$1.

Model	Δ_{GSM8K} (pp)	Δ_{MMLU} (pp)	RE
Llama-3.1-8B†	-87.0	-15.9	$5.47\times$
Qwen3-8B†	-93.6	-52.0	$1.80\times$
Qwen2.5-14B‡	-28.2	-6.35	$4.43\times$
Phi-3-medium-14B‡	-26.6	-10.35	$2.57\times$
DeepSeek-R1-7B	-23.8	-9.5	$2.51\times$
Qwen2.5-7B	-11.9	-9.5	$1.26\times$
Gemma-2-9B	-3.4	-5.2	$0.65\times$

†Collapse models: GSM8K dropped to 0%. 95% CIs in Appendix A.22. ‡MMLU evaluated on 50 questions per subject (n=2,850); remaining models use full MMLU (n=14,042). Bootstrap 95% CIs: three cliff-regime models have $p < 0.001$ after Bonferroni correction ($\alpha = 0.01$, m=5); Qwen2.5-7B is significant uncorrected ($p = 0.016$) but its 99% CI $[0.94, 1.59]$ contains 1.

devastated, consistent with its degenerate output of repeated exclamation marks rather than the structured cross-lingual tokens Llama produces (Appendix A.12).

Only Gemma-2 inverts the pattern (RE $= 0.65$), but within-MMLU subject analysis—stratifying reasoning-intensive vs. knowledge-intensive subjects—reverses Gemma-2's RE to $1.43\times$, confirming reasoning specificity in four of six models (mean within-MMLU RE $= 1.72\times$; Appendix A.38). ARC-Challenge and HumanEval provide additional cross-benchmark evidence (Appendix A.20, A.24).

4.3 Damage Accumulates Through Reasoning Chains

The **flip rate**—the fraction of FP16-correct questions that become incorrect under W3A16—reveals per-question instability hidden by aggregate accuracy. Cliff-regime flip rates reach 29.1–31.1%; even the mildest-affected Qwen2.5-7B flips 15.7% of questions despite only an 11.9 pp aggregate drop (Table 9, Appendix A.4). Critically, in 3 of 5 models longer FP16 chains are significantly more likely to flip ($p < 0.01$; Spearman $\rho = 0.12$–0.16), consistent with per-token error accumulation (generation length is the dominant driver after controlling for chain steps; §5.2); the two highest-flip models show no length correlation, suggesting damage saturation beyond a severity threshold (Figure 3). A simple geometric error-accumulation model fits the three length-correlated models well (binned $R^2 = 0.65$–0.87, per-token error rate $\varepsilon = 0.6$–1.7×10^{-3}), while the two saturated models (DeepSeek, Qwen2.5-14B) exceed the accumulation regime—their flip rates near 30% approach the ceiling of available FP16-correct questions, masking further length dependence. Logistic regression corroborates the pattern (odds ratios 1.68–$2.54\times$ per 100 tokens; Appendix A.4).

4.4 Error Structure, Not Magnitude, Determines Damage

The preceding sections establish *what* degrades (reasoning over knowledge), *where* damage concentrates (longer chains), and how steeply it onsets (the W4→W3 cliff). A mechanistic question remains: is the cliff driven by the *magnitude* of quantization error, or by its *structure*? We isolate these factors via **perturbation replay**: injecting the exact per-layer GPTQ rounding error $\Delta W = W_{\text{q}} - W_{\text{fp16}}$ into FP16 weights, and comparing the resulting accuracy to that produced by i.i.d. Gaussian noise matched in per-layer standard deviation (Eq. 2; Table 3).

Full-scale replay (c=1.0). For Gemma-2, the structured perturbation closely tracks the actual cliff: 80.6% (-3.4 pp from the 84.0% FP16 baseline), with zero truncation. Replacing this structured error with Gaussian noise of identical magnitude produces 3.3–5.6× more damage (mean $4.5\times$, three seeds): three seeds yield 72.8%, 68.6%, and 65.0% (mean $68.8\pm3.9\%$, $\Delta{=}-15.2$ pp).[1] Equivalently, the structured GPTQ error accounts for only $\sim$22% of the damage caused by random noise of the same per-layer standard deviation.

Phi-3-medium-14B shows a second profile. Under structured replay, accuracy drops to 57.4% (-26.8 pp from the 84.2% 0-shot baseline), closely reproducing the actual GPTQ W3A16 cliff (-26.6 pp, Table 1) within 0.2 pp, with only 1.2% truncation. Gaussian noise of matched magnitude causes complete collapse (0.2%, $\Delta = -84$ pp), yielding a structured-to-Gaussian damage ratio of $3.1\times$ ($\pm0.1\times$, two seeds; Table 3). This collapse is consistent across seeds (seed 42 and seed 123 both produce 0.2%; per-seed ratios $3.13\times$ and $3.04\times$).

DeepSeek-R1-7B reveals a qualitatively different pattern. Under structured replay at 8192 tokens, accuracy drops to 70.2% (-19.6 pp from the 89.8% baseline), with 30% truncation versus 2.2% at FP16—genuine reasoning damage, not a truncation artifact. The 4.2 pp gap between structured replay (-19.6) and the actual GPTQ cliff (-23.8 pp, Table 1) likely reflects the longer generation budget (8192 vs. 4096 tokens), different evaluation subsets (n=500 vs. n=450 common subset), and quantization effects beyond per-layer rounding error (e.g., scale/zero-point discretization). Gaussian replay at c=1.0 causes near-total generation collapse: 90–99% of responses exceed the token limit across four seeds (seed 42: 26.2%, seed 123: 11.2%, seed 456: 5.0%, seed 789: 19.8%; mean $15.6\pm9.3\%$, $\Delta = -74.2$ pp; 95% CI $[0.7\%, 30.4\%]$, t_3). The resulting ratio ($3.8 \pm 0.5\times$, four seeds) aligns with Gemma-2 and Phi-3, though DeepSeek's cross-seed variance (SD $=$ 9.3 pp, 95% CI spanning nearly the full accuracy range) is substantially higher than Gemma-2's (SD $=$ 3.9 pp, three seeds), likely because long reasoning chains (3755 ± 1022 tokens) amplify stochastic noise accumulation and the high truncation rate ($\sim$$90\%$) makes accuracy sensitive to where generation is cut (Appendix A.30). The qualitative contrast—structured noise degrades reasoning while Gaussian noise destroys generation entirely—further confirms that error structure, not magnitude, determines the failure mode.

Across all three models at c=1.0, Gaussian noise of matched per-layer standard deviation causes 3–5× more damage than the structured GPTQ error: Gemma-2 $4.5\times$ ($\pm1.2\times$, three seeds), DeepSeek $3.8\times$ ($\pm0.5\times$, four seeds), Phi-3 $3.1\times$ ($\pm0.1\times$, two seeds) (Table 3). GPTQ's calibration-based rounding preserves weight-space structure that shields reasoning from most of the perturbation's potential harm. A supplementary half-scale experiment (c=0.5, n=300) shows directionally consistent but individually non-significant effects (structured $+1.0$ pp, Gaussian -1.7 pp on DeepSeek; Table 3). Dose-response curves (Figure 2) and module-level decomposition (Appendix A.27) provide further evidence.

[1] Seed 0 was the initial exploratory run; seeds 42 and 123 were added for variance estimation. Per-seed ratios range from $3.3\times$ to $5.6\times$, confirming that the mean $4.5\times$ is robust across noise realizations. Seed 789 (n=300) excluded for sample-size consistency; including it yields mean 69.2 ± 3.3%, qualitatively unchanged.

Table 3: Perturbation replay: GSM8K accuracy under structured ΔW and Gaussian noise at two scales. At c=1.0, Gaussian noise causes 3–5× more damage than structured error across all three models (Gemma-2 $4.5\times$, DeepSeek $3.8\times$, Phi-3 $3.1\times$; all with ±SD from two–four seeds). At c=0.5, structured and Gaussian noise show opposing trends (neither individually significant at n=300), consistent with the c=1.0 pattern.

Model	Condition	Acc (%)	Δ (pp)	Trunc (%)
Full-scale replay (c=1.0, n=500)				
	FP16 baseline	89.8	—	2.2
DeepSeek-R1-7B	Structured ΔW	70.2	−19.6	30.0
	Gaussian (4 seeds)	15.6±9.3	−74.2	90–99
	FP16 baseline†	84.0	—	0.0
Gemma-2-9B	Structured ΔW	80.6	−3.4	0.0
	Gaussian (3 seeds)	68.8±3.9	−15.2	0.0
	FP16 baseline¶	84.2	—	0.0
Phi-3-medium-14B	Structured ΔW	57.4	−26.8	1.2
	Gaussian	0.2	−84.0	0.0
Half-scale replay (c=0.5, n=300)				
	FP16 baseline	90.0	—	1.7
DeepSeek-R1-7B	Structured ΔW	91.0	+1.0	4.3
	Gaussian	88.3	−1.7	6.7

c=1.0: n=500. ¶0-shot. †Internal baseline (420/500). DeepSeek structured at 8192 tokens, Gaussian at 4096; FP16 baseline (89.8%) uses chatml, 8192 tokens (cf. Table 1: 78.0%, 0-shot, 4096). c=0.5: n=300, seed 42. Details in Appendix A.27.

5 Analysis

A case study on DeepSeek-R1-7B reveals 23× more self-doubt markers ("wait, no") under W3A16, suggesting that quantization amplifies hesitation and self-correction attempts (Appendix A.1).

5.1 Cliff Severity Scales with Reasoning Chain Length

Stratifying DeepSeek-R1-7B's MATH-500 results by difficulty (Table 7, Appendix A.2), the W3A16 cliff scales sharply: from −14 pp at Level 1 (mean 1,343 tokens) to −52 pp at Level 5 (mean 4,482 tokens). At the hardest level, 76% of FP16-correct answers are destroyed. W4A16 shows no comparable gradient. This co-variation of cliff severity with difficulty and generation length supports error accumulation along the reasoning chain, and explains why the cliff is more severe on MATH-500 than GSM8K: competition-level problems elicit longer chains. A second model (Qwen2.5-14B) replicates the gradient (Appendix A.15).

5.2 Disentangling Token Length from Reasoning Depth

Chain-step count and generation length are highly collinear (Spearman r=0.73–0.87 across five models: Qwen2.5-7B 0.74, Qwen2.5-14B 0.73, Phi-3 0.84, Gemma-2 0.85, DeepSeek 0.87), so raw correlations cannot separate the two drivers. We compute partial correlations to disentangle them.

After controlling for chain-step count, generation length remains a significant positive predictor of flip probability in all five models (r=0.07–0.19, all p<0.02; Table 4). While modest in magnitude (R^2=0.5–3.6%), the direction is consistent across all five models, suggesting a systematic rather than spurious association. The reverse partial—chain steps controlling for generation length—is non-significant or *significantly negative* in every case: Qwen2.5-14B (r=−0.110, p=0.0001) and Gemma-2 (r=−0.112, p=0.023) show that, at fixed token count, more chain steps *reduce* flip probability, suggesting that shorter per-step reasoning carries denser critical information. In all five models, discrete reasoning depth adds no independent risk beyond its effect on token count. Qwen2.5-14B illustrates a suppressor effect: its raw Spearman correlation between generation length and flip

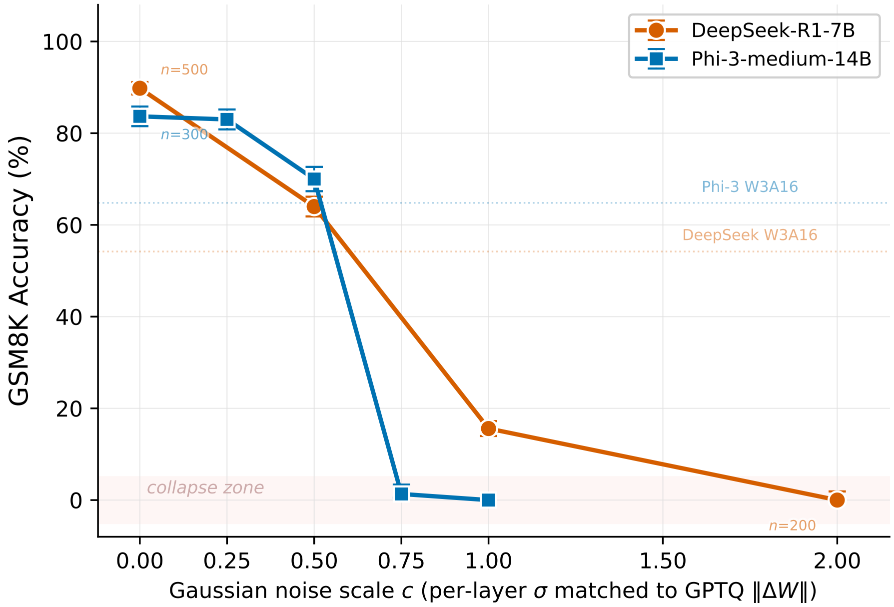

Figure 2: Dose-response: GSM8K accuracy under Gaussian noise scaled to c times the per-layer GPTQ error standard deviation, with 95% Wilson CIs. Phi-3 collapses sharply between c=0.5 and c=0.75; DeepSeek degrades more gradually but reaches zero by c=2.0. Dotted lines: actual W3A16 accuracy. Details in Appendix A.28.

Table 4: Partial correlations with flip probability. Controlling for chain steps, generation length remains a significant predictor in all five models; the reverse partial is zero or negative. DeepSeek probing uses 8192-token budget (matching its perturbation replay protocol in Table 3); all other models use 4096.

	r(gen_len, flip | steps)		r(steps, flip | gen_len)	
Model	r	p	r	p
Qwen2.5-7B	0.155	<0.001	−0.034	0.237
Qwen2.5-14B	0.074	0.009	−0.110	0.0001
Phi-3-14B	0.113	0.016	−0.010	0.829
Gemma-2-9B	0.193	<0.001	−0.112	0.023
DeepSeek-R1-7B	0.133	0.005	−0.083	0.078

rate is near zero (r=−0.027, p=0.34), yet the partial correlation after controlling for chain steps is significant (r=0.074, p=0.009).

Token-length quartile analysis confirms the pattern: Q4 (longest outputs) flip rates exceed Q1 (shortest) in four of five models—Qwen2.5-7B 7.0%→24.7%, Phi-3 20.9%→42.5% (both strictly monotonic), DeepSeek 19.5%→40.2%, and Gemma-2 0.9%→14.4%. Qwen2.5-14B is the exception (33.8%→32.3%, Q4<Q1), consistent with the suppressor pattern noted above: its high baseline flip rate (31%) compresses the quartile gradient. The dominant statistical predictor of quantization-induced reasoning degradation is thus cumulative token-level error, not discrete reasoning steps per se; multi-step problems are vulnerable because they produce more tokens. The direction of association is unambiguous: generation length is measured from FP16 outputs (a proxy for problem difficulty), which is causally prior to the quantization-induced flip; reverse causation is therefore excluded by design.

5.3 Architecture-Dependent Collapse Modes

At c=2.0 perturbation, both DeepSeek and Gemma-2 reach near-zero accuracy but through distinct pathways: DeepSeek via *generation collapse* (87.5% truncation, mean 3,906 tokens) and Gemma-2

Table 5: Cross-method comparison at W3A16 on GSM8K (N=500). DeepSeek evaluated with 8-shot CoT; Gemma-2 with 0-shot (matching each model's main evaluation protocol). Δ relative to matched FP16 baseline. The cliff severity is strongly method-dependent for DeepSeek but compressed for mild-regime Gemma-2.

Method	Precision	Acc (%)	Δ
DeepSeek-R1-7B (8-shot CoT)			
FP16 (baseline)	16-bit	87.2[§]	—
GPTQ (gs=128)	3-bit	54.2	−33.0 pp
RTN (gs=128)	3-bit	62.6	−24.6 pp
HQQ	3-bit	84.2	−3.0 pp
Gemma-2-9B (0-shot)			
FP16 (baseline)	16-bit	83.6[¶]	—
GPTQ (gs=128)	3-bit	80.2	−3.4 pp
RTN (gs=128)	3-bit	82.2	−1.4 pp

[§] 8-shot, chatml template, 4096-token budget; matched to GPTQ/RTN/HQQ protocol. Higher than Table 1 (0-shot, N=450) due to in-context exemplars and easier first-500 subset. [¶] 0-shot, matching Table 1 protocol. DeepSeek truncation: RTN 31.0%, HQQ 20.6%; Gemma-2 truncation 0% across methods.

via *reasoning collapse* (0% truncation, normal-length but wrong outputs). Standard output-length monitoring would detect DeepSeek's failure but miss Gemma-2's entirely. This divergence may reflect DeepSeek's reasoning-distillation training, which creates an additional vulnerability surface—the self-correction mechanism itself—that amplifies damage beyond what weight perturbation alone would produce (Appendix A.1). Attention entropy is confounded by generation length ($r = 0.939$; Appendix A.32).

Group-size mitigation. Finer GPTQ grouping (gs=64) recovers cliff-regime models (DeepSeek $+30.8$ pp, Qwen2.5-14B $+10.6$ pp) but not mild-regime Qwen2.5-7B (-1.4 pp), suggesting scale-dependent mitigation within GPTQ (Table 11, Appendix A.6).

Layer-level localization. Mid layers are most vulnerable to perturbation replay across architectures. DeepSeek shows a graded sensitivity profile: mid layers (10–18) sustain the largest damage (-5.7 pp), back layers (19–27) moderate (-3.3 pp), and front layers (0–9) are most resilient (-1.3 pp). Gemma-2 confirms mid-layer dominance (-4.0 pp, layers 14–27) with near-immune back layers ($+0.3$ pp, layers 28–41) and moderate front damage (-2.0 pp, layers 0–13). Phi-3-medium-14B (40 layers) adds a third confirmation: mid layers (14–26) sustain the largest damage (-5.3 pp), back layers (27–39) moderate (-3.7 pp), and front layers (0–13) are most resilient (-1.0 pp), closely paralleling DeepSeek's graded profile. The cross-model consensus across all three architectures—front-resilient, mid-vulnerable—suggests that intermediate transformer layers, where multi-step reasoning chains are assembled, bear the brunt of quantization-induced error. Back-layer sensitivity diverges across architectures (DeepSeek -3.3 pp, Phi-3 -3.7 pp vs. Gemma-2 $+0.3$ pp), possibly reflecting depth-ratio differences (28/40 vs. 42 layers). Non-additivity differs in direction: Gemma-2's group sum (-5.7 pp) exceeds its full-model effect (-3.4 pp), indicating partial compensation across layers, while DeepSeek's group sum (-10.3 pp) and Phi-3's (-10.0 pp) fall short of their full-model effects (-19.6 pp and -26.8 pp respectively), consistent with error propagation and compounding—perturbations in early layers are amplified as they feed through deeper layers. We note that layer-group and full-model experiments used different generation budgets (4096 vs. 8192 tokens); the non-additivity magnitude should be interpreted with this caveat (Appendix A.34).

5.4 Cross-Method Analysis: GPTQ vs. RTN vs. HQQ

To disentangle precision from algorithm, we evaluate DeepSeek-R1-7B at 3-bit using two alternatives: RTN (round-to-nearest, no calibration) and HQQ [Badri and Shaji, 2023] (half-quadratic optimization of scale/zero-point without calibration data). We additionally test RTN on mild-regime Gemma-2 to assess whether tier classification is method-independent.

On DeepSeek-R1-7B, the three methods span a 30.0 pp range at identical 3-bit precision ($N{=}500$). GPTQ (54.2%) is worst, RTN (62.6%) partially recovers, and HQQ (84.2%) shows minimal degradation (-3.0 pp). Counterintuitively, GPTQ produces *worse* reasoning than naive RTN; one explanation is that its Hessian-based objective redistributes residual error in ways that disproportionately affect multi-step reasoning. Critically, the cliff is not GPTQ-specific: RTN, which applies no calibration or error compensation whatsoever, still exhibits a substantial cliff (-24.6 pp). Gemma-2 replicates the method ordering (RTN $>$ GPTQ) but with compressed severity: both methods stay within 3.4 pp of the 83.6% FP16 baseline, confirming that the mild-regime classification is method-independent. The 87.2% FP16 baseline here (8-shot CoT) exceeds Table 1's 78.0% (0-shot) by 9.2 pp, yet GPTQ remains at 54.2% in both protocols. W3A16 GPTQ does not appear to benefit from few-shot exemplars, consistent with quantization potentially interfering with the in-context learning pathway, though a floor effect at this severity level cannot be ruled out. The methods thus establish that 3-bit precision creates a reasoning vulnerability whose severity is both method- and architecture-dependent—GPTQ most severely on cliff-regime models, while mild-regime models remain resilient across methods. This does not contradict the replay analysis (§4.4): replay holds magnitude constant (GPTQ's structure is protective per unit norm), while cross-method comparisons involve different total error budgets.

6 Discussion

Structured error as a protective mechanism. The Gaussian/ΔW damage ratio—Gemma-2 $4.5\times$ ($\pm 1.2\times$, three seeds), DeepSeek 3.0–$4.5\times$ (mean $3.8\times$, four seeds; 95% CI of accuracy: $[0.7\%, 30.4\%]$), Phi-3 $3.1\times$ ($\pm 0.1\times$, two seeds)—shows that GPTQ error retains protective structure at matched magnitude. GPTQ's calibration-based rounding provides a dual benefit: *lower damage* per unit perturbation norm and *reproducible outcomes* (<0.3 pp cross-seed variance vs. 10–20 pp for Gaussian noise)—though this per-norm property does not prevent GPTQ from being worst overall at 3-bit (§5.4). Partial perturbation at module and head level fails to reproduce the full cliff (Appendix A.27), implying effective mitigation requires uniform precision above the threshold or quantization objectives that explicitly preserve chain-of-thought coherence.

The cliff as a coherence threshold. The converging evidence—three-tier spectrum, difficulty gradient, and token-length–flip correlation (§5.2)—suggests quantization errors accumulate along reasoning chains until crossing a model-specific coherence threshold, with architecture determining where that threshold lies. RE's cross-modality validity is confirmed by ARC-Challenge and within-MMLU subject analysis (Appendix A.20, A.38). Broader impact considerations appear in Appendix A.35.

7 Conclusion

We presented the first systematic study of how aggressive weight quantization degrades multi-step reasoning across 7 models, five families, three precisions, and five benchmarks. W3A16 GPTQ produces a three-tier spectrum (collapse/cliff/mild) determined by architecture, not scale; the cliff selectively targets reasoning (RE >1 in six of seven models, three significant after Bonferroni correction); per-question flip rates up to 31% reveal instability hidden by aggregate accuracy; partial-correlation analysis confirms that cumulative token length, not discrete reasoning steps, is the dominant predictor of flip probability; and perturbation replay confirms that error structure, not magnitude, determines damage severity.

Practical implications. W4A16 is safe for reasoning deployment (all nine W4A16 McNemar tests non-significant after Bonferroni correction; smallest uncorrected $p = 0.027$, Llama $\times$ MATH-500); W3A16 is dangerous even when aggregate accuracy survives (flip rates $\geq 29\%$). We recommend RE combined with per-sample flip rate as a standard reasoning-safety evaluation protocol for quantized models, complementing perplexity-based assessments which cannot distinguish cliff from mild regimes ($+33.8\%$ ppl vs. -23.8 pp GSM8K on DeepSeek).

Limitations. Our W3A16 results characterize GPTQ specifically; other methods (RTN, HQQ) show different cliff severities (§5.4), so the three-tier spectrum may not generalize to all quantization algorithms. The 7–14B scale range leaves open whether the cliff persists at 70B+, where greater weight redundancy may provide natural protection. All evaluations use a single seed; stochastic decoding could alter flip rates. Our perturbation replay and layer-level analysis provide mechanistic evidence for *how* quantization damages reasoning (error structure and mid-layer concentration), but a full causal account of chain-of-thought disruption remains open. Additional caveats are detailed in Appendix A.36.

Future work. Key directions include scale extension to 70B+, universality testing via QuIP#/AQLM, and reasoning-aware quantization objectives (Appendix A.40).

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Table 6: Self-doubt cascade markers on GSM8K (mean count per question). FP16 ($N = 1319$) vs. W3A16 ($N = 500$). These cascade patterns are nearly absent in FP16 and cannot be explained by the $1.8\times$ generation-length difference.

Marker	FP16	W3A16	Ratio
`wait_no`	2.4	54.9	23×
`no_no`	1.0	30.8	31×

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A Extended Results

A.1 Self-Doubt Case Study

We analyze DeepSeek-R1-Distill-Qwen-7B, the only reasoning-distilled model with explicit self-correction traces. Table 6 reports self-doubt marker frequencies: W3A16 produces $23\times$ more "wait, no" and $31\times$ more "no, no" cascade patterns than FP16—markers that are essentially absent in FP16 outputs (≤ 2.4 per question) but appear dozens of times under W3A16. These ratios far exceed the $1.8\times$ generation-length difference, confirming they reflect quantization-induced reasoning instability rather than a length artifact. The effect concentrates on flipped questions (FP16 correct $\rightarrow$ W3A16 incorrect): mean doubt-marker differential is 9.4 vs. 3.3 on non-flipped questions ($2.8\times$ gap), indicating self-doubt accumulation is associated with answer degradation.[2] Error taxonomy (Table 10, Appendix A.5) reveals repetitive loops as the dominant W3A16 failure mode (85.4% vs. 16.9% FP16, $5.1\times$), while arithmetic errors *decrease* (5.3% vs. 15.9%)—the model is trapped in self-doubt loops before reaching computation. However, 56.9% of W3A16 errors hit the token limit; precise frequencies should be interpreted with this truncation caveat.

A.2 Difficulty Stratification Detail

A.3 Model Details

A.4 Flip Rate Detail

A.5 Error Taxonomy Detail

A.6 Group-Size Mitigation Detail

For severe-cliff DeepSeek-R1-7B, gs=64 produces dramatic recovery ($+30.8$ pp), nearly matching its FP16 matched-subset baseline (85.0% vs. 90.0%), consistent with the error-accumulation mechanism from §5.1. Qwen2.5-14B also benefits ($+10.6$ pp), with non-truncated accuracy (71.7%) *lower* than overall (72.2%), ruling out truncation as the explanation. However, Qwen2.5-7B shows no response (-1.4 pp), indicating mild-regime degradation is not driven by per-group granularity. We recommend evaluating both group sizes on a small held-out reasoning set before committing to the $\sim 2\times$ scale-parameter overhead of gs=64.

[2] The behavioral analyses use the full W3A16 evaluation (N=500) matched against FP16 (N=1319) by `question_idx`, distinct from the 450-question three-precision subset used for accuracy comparisons.

Table 7: DeepSeek-R1-7B accuracy (%) on MATH-500 by difficulty (n=496). The W3A16 cliff scales with difficulty—from -14 pp at Level 1 to -52 pp at Level 5—mirroring mean generation length. Difficulty from Hendrycks et al. [2021].‡

Level	n	FP16	W4A16	W3A16	Δ_{cliff}	Mean tokens
1	43	74.4	86.0	60.5	−13.9	1,343
2	90	78.9	83.3	65.6	−13.3	1,922
3	104	82.7	80.8	46.2	−36.5	2,223
4	127	79.5	77.2	40.9	−38.6	3,068
5	132	68.2	70.5	16.7	−51.5	4,482

‡Accuracy scored with the initial extraction pipeline; the improved extractor (Appendix A.13) raises absolute accuracy by $\sim$3 pp uniformly, preserving the gradient.

Table 8: The 7 models evaluated in this study.

Model	Family	Params	Source
Llama-3.1-8B-Instruct	Llama-3	8.0B	Meta
Qwen2.5-7B-Instruct	Qwen2	7.6B	Alibaba
Qwen2.5-14B-Instruct	Qwen2	14.7B	Alibaba
Qwen3-8B	Qwen3	8.2B	Alibaba
Phi-3-medium-14B-instruct	Phi-3	14.0B	Microsoft
Gemma-2-9B-it	Gemma-2	9.2B	Google
DeepSeek-R1-Distill-Qwen-7B	Qwen2†	7.6B	DeepSeek

†Reasoning-distilled from a DeepSeek-R1 teacher; architecturally Qwen2-based.

A.7 Per-Head Profile Shift Data

Table 12 reports the five KV head groups with the largest 4D profile shift magnitude ($\delta^{(\ell,h)}$, the L_2 norm of the z-scored shift vector) under FP16→W3A16 quantization for DeepSeek-R1-Distill-Qwen-7B (28 layers × 4 KV heads = 112 groups total). Values are extracted from `results/shift_analysis/deepseek_shifts.json`.

A.8 Perturbation Replay Calibration

The perturbation replay protocol uses the *exact* quantization error as the perturbation signal, as described in §3.5. For each layer ℓ, we compute the weight difference between the dequantized W3A16 GPTQ model and the original FP16 model:

$$\Delta W^{(\ell)} = W^{(\ell)}_{\text{dequant(W3)}} - W^{(\ell)}_{\text{FP16}} \tag{3}$$

for both the K and V projection matrices. At inference time, for each selected head subset, we add $F\big(x,\, \Delta W^{(\ell)}_{\text{subset}}\big)$ to the FP16 model's K/V activations via a forward hook, effectively replaying the quantization-induced weight error on exactly those heads.

Table 13 reports the L_2 norm of ΔW (profile shift magnitude in 4D space) for target and control heads, confirming that the two groups differ in functional disruption rather than raw perturbation magnitude.

A.9 Error Taxonomy Classification Rules

Table 10 categorizes errors into five non-exclusive types. Classification uses the following detection rules applied to each model response:

- **Repetitive loop**: A substring of ≥20 tokens that repeats ≥3 consecutive times within the response. Detection: sliding window over tokenized output, checking for exact n-gram repetition with $n \in \{20, 50, 100\}$.

[3]An earlier experiment with different eval settings yielded -5.2 pp; the matched-setting rerun (`exp_gemma2_structured_500q`) gives -3.4 pp.

Table 9: GSM8K per-sample flip rate (FP16 correct → W3A16 incorrect). Qwen2.5 models use the full GSM8K test set (n=1,319) while other models use subsets of 450–500 questions; flip rates are reported as proportions and remain comparable across sample sizes.

Model	n	Flip rate (%)	Δ_{cliff} (pp)
DeepSeek-R1-7B	450	30.8	−23.8
Qwen2.5-14B	1319	31.1	−28.2
Phi-3-medium-14B	500	29.1	−26.6
Qwen2.5-7B	1319	15.7	−11.9
Gemma-2-9B	500	8.1	−3.4

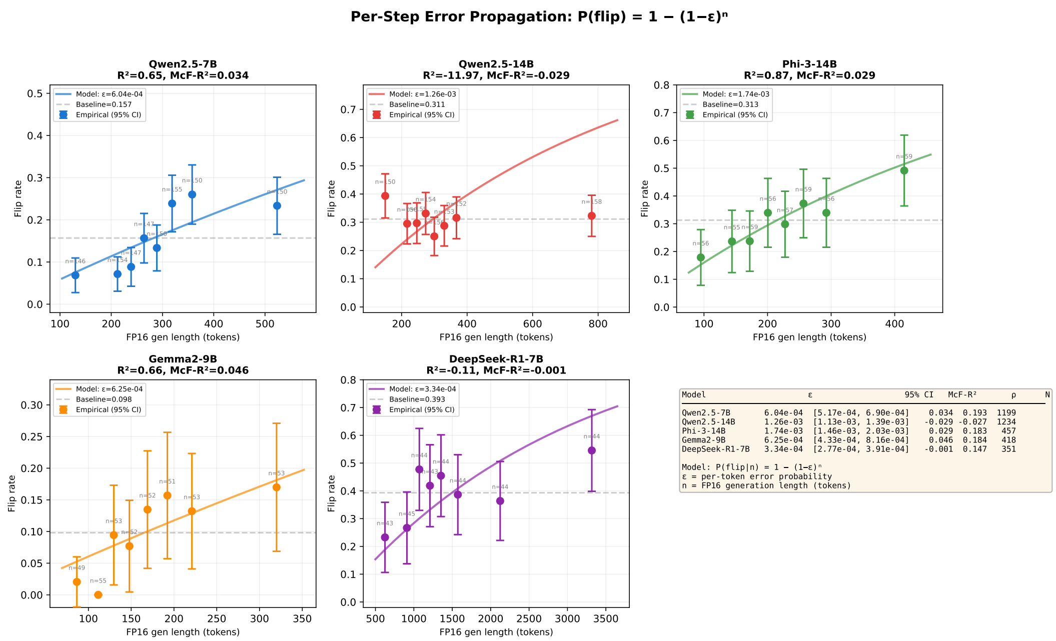

Figure 3: Per-step error propagation model $P(\text{flip} \mid n) = 1 - (1 - \varepsilon)^n$ fitted to five models. Each subplot shows empirical flip rates (binned by FP16 generation length, 95% Wilson CIs) and the geometric model fit. Three models show good fits (Qwen2.5-7B R^2=0.65, Phi-3 R^2=0.87, Gemma-2 R^2=0.66), consistent with per-token error accumulation ($\varepsilon = 0.6$–1.7×10^{-3}). Qwen2.5-14B and DeepSeek-R1 show no length dependence ($R^2 < 0$), consistent with damage saturation at their higher flip-rate baselines (≈0.31). Summary table (bottom right) reports per-token error rates with 95% CIs.

- **Hit token limit**: The response reaches the `max_new_tokens` ceiling (4096 tokens) without a natural stopping point. Detection: `len(output_tokens) >= max_new_tokens`.
- **Answer corrupted by loop**: A correct intermediate answer (matching the ground truth) appears in the response, but the final extracted answer (after the last `####` delimiter) is incorrect. Detection: regex search for the ground-truth number in the response body, combined with incorrect final-answer extraction.
- **Self-contradiction**: The response contains explicit reversal patterns where the model negates a previous statement. Detection: co-occurrence of an affirmative claim and its negation within a 200-token window, identified by patterns such as "the answer is X . . . no, the answer is Y" where $X \neq Y$.
- **Arithmetic error**: The final answer is wrong due to a computational mistake in an otherwise coherent reasoning chain (no loops, no contradictions). Detection: manual verification on a random subset ($n = 50$), confirmed by checking that intermediate steps follow logically but a specific calculation is incorrect.

Table 10: Error taxonomy on GSM8K for DeepSeek-R1-7B: W3A16 vs. FP16 ($N = 229$ errors each). Categories are non-exclusive and may overlap with truncation-induced artifacts (56.9% of W3A16 errors hit the token limit). Classification rules in Appendix A.9.

Error type	**FP16 (%)**	**W3A16 (%)**	**Ratio**
Repetitive loop	16.9	85.4	5.1×
Hit token limit	13.3	56.9	4.3×
Answer corrupted by loop	3.5	27.5	7.8×
Self-contradiction	6.5	13.5	2.1×
Arithmetic error	15.9	5.3	0.3×

Table 11: GSM8K accuracy (%) under GPTQ 3-bit with group size 128 vs. 64. Mitigation effectiveness is model-dependent: DeepSeek recovers $+30.8$ pp, Qwen2.5-14B gains $+10.6$ pp, while Qwen2.5-7B is unaffected (-1.4 pp).

Model	gs=128	gs=64	Δ (pp)	Regime
DeepSeek-R1-7B	$54.2^\dagger$	85.0	$+30.8$	Cliff (severe)
Qwen2.5-7B	$76.4^\S$	75.0	-1.4	Mild
Qwen2.5-14B	$72.2^\ddagger$	82.8	$+10.6$	Cliff (moderate)

†Truncation rate 24.4% at gs=128. §gs=128 re-evaluated with matched settings (8-shot, 500 questions); truncation rate 0.8%. ‡gs=128 truncation rate 8.8%; non-truncated accuracy is 71.7%, *lower* than overall 72.2%—truncated samples have higher accuracy (77.3%), indicating genuine reasoning failure rather than generation-budget exhaustion.

Categories are non-exclusive: a single response can be classified under multiple types (e.g., a response may contain both a repetitive loop and hit the token limit).

A.10 Self-Doubt Marker Regular Expressions

The self-doubt markers reported in Table 6 are detected using the following case-insensitive regular expressions applied to the raw text output:

- `wait_no`: Matches hesitation-then-negation patterns.

  ```
  r"[Ww]ait[,.\s]*(?:no|but|actually|hold on|let me)"
  ```

- `no_no`: Matches repeated negation indicating self-correction.

  ```
  r"[Nn]o[,.\s]*no[,.\s]*(?:that's|this|I|wait)"
  ```

- `direction_changes`: Matches points where the model abandons a reasoning path and restarts. Detected by the conjunction of a conclusion marker followed by a reversal:

  ```
  r"(?:so the answer is|therefore|thus|hence).*?"
  r"(?:wait|but|actually|no|hmm|let me reconsider)"
  ```

  Applied within a 150-token sliding window.

Each match is counted once per occurrence (not per question), then aggregated to per-question frequencies by dividing total matches by the number of questions.

A.11 Generation Length Distribution

Table 14 reports the distribution of generation lengths (in tokens) for DeepSeek-R1-Distill-Qwen-7B on GSM8K under FP16 and W3A16.

Table 12: Top-5 KV head groups by 4D profile shift magnitude (FP16→W3A16) for DeepSeek-R1-Distill-Qwen-7B. Each dimension is z-scored against FP16 cross-head standard deviation before computing the L_2 norm. Heads are sorted by total shift δ.

Head	Sink Δz	Entropy Δz	Gini Δz	Bias Δz	δ	Dominant shift
L16_KV0	$+2.70$	-2.50	$+1.69$	-0.98	4.16	sink$\uparrow\uparrow$, entropy$\downarrow\downarrow$
L23_KV0	-2.62	$+2.61$	-1.15	$+1.41$	4.12	sink$\downarrow\downarrow$, entropy$\uparrow\uparrow$
L18_KV2	$+1.71$	-1.37	$+0.69$	-3.15	3.90	bias$\downarrow\downarrow\downarrow$
L3_KV1	-0.52	$+1.46$	-3.32	-0.39	3.68	gini$\downarrow\downarrow\downarrow$
L12_KV3	$+2.15$	-1.72	$+0.86$	-2.16	3.61	sink$\uparrow\uparrow$, bias$\downarrow\downarrow$

Representative raw values: L16_KV0 sink $0.274 \to 0.699$ ($\Delta = +0.425$), entropy $4.27 \to 2.03$ ($\Delta = -2.24$); L23_KV0 sink $0.676 \to 0.461$ ($\Delta = -0.214$), entropy $2.12 \to 3.05$ ($\Delta = +0.93$). Summary across all 112 heads: 12 heads exceed $|z| > 2$ on at least one dimension (entropy: 11, sink: 6, gini: 4, with 7 heads significant on multiple dimensions). The top-5 heads span layers 3–23, with no single dimension dominating: sink, entropy, gini, and bias each appear as the primary shift in at least one head.

Table 13: Profile shift magnitude (4D L_2 norm of z-scored shift, 2-question calibration) for target and control heads used in the perturbation replay experiment. Target heads are the five most disrupted; control heads are the five least disrupted. Profile shift values are computed from the 2-question calibration subset used for perturbation replay head selection; these differ from the full-dataset profiles in Table 12 due to smaller sample size, which inflates per-head z-scores. The distributed-damage conclusion (§5) is robust to head selection criteria, as neither target nor control subsets reproduce full W3A16 degradation.

Head	Profile shift δ	Role
Target heads (top-5 profile shift)		
L4_KV0	4.16	shallow, sink$\uparrow$
L4_KV2	3.90	shallow, sink$\uparrow$
L4_KV3	3.68	shallow, sink$\uparrow$
L23_KV0	29.51	deep, entropy$\uparrow$
L24_KV2	24.65	deep, entropy$\uparrow$
Control heads (minimal profile shift)		
L5_KV3	2.44	minimal
L7_KV1	2.50	minimal
L25_KV2	3.53	minimal
L23_KV3	3.53	minimal
L21_KV1	2.50	minimal

The large gap in profile shift magnitude between target (δ up to 29.5) and control ($\delta \leq 3.5$) heads confirms that head selection is based on functional disruption severity. Despite this difference, the perturbation replay experiment (Table 3) shows comparable accuracy degradation across conditions, supporting the distributed-damage interpretation. Note that while the 2-question calibration is an experimental limitation, the distributed-damage conclusion does not depend on specific head selection: the `all_weights` condition—which injects perturbation into *every* linear layer regardless of head selection—produces only -19.2 pp (DeepSeek) and -3.4 pp (Gemma-2, matched-setting evaluation[3]), closely tracking the true GPTQ cliff (-23.8 pp and -3.4 pp respectively).

A.12 Degenerate Output Examples

We illustrate the three distinct degenerate output patterns observed under W3A16 quantization.

Qwen3-8B: Exclamation loop. Qwen3-8B under W3A16 GPTQ produces outputs dominated by repeated exclamation marks, with no mathematical content:

```
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
[continues for 4096 tokens]
```

Table 14: Generation length distribution (tokens) for DeepSeek-R1-7B on GSM8K. W3A16 produces substantially longer outputs, with a $2.9\times$ increase in mean length and 56.9% of responses reaching the 4096-token ceiling.

Precision	N	Mean	Median	P25	P75	Max
FP16	1319	871	642	384	1108	4096
W3A16	500	2547	2831	1204	3892	4096

FP16: 20.8% of responses reach the 4096-token limit. W3A16: 56.9% reach the limit. The $2.9\times$ increase in mean generation length reflects the self-doubt loop phenomenon (§A.1), where the model repeatedly revises its reasoning chain before converging (or failing to converge) on an answer.

This pattern accounts for the vast majority of Qwen3's W3A16 outputs (100% truncation rate). The model's weight corruption is so severe that it cannot produce coherent language tokens.

Llama-3.1-8B: Cross-lingual token collapse. Llama under W3A16 begins with fragmentary English before collapsing into repetitive Japanese characters (hiragana), cycling through a small token set indefinitely:

```
The 1 was a toN to of the next[JP][JP][JP][JP],
1[JP][JP][JP][JP][JP][JP],1[JP][JP][JP][JP][JP]
[JP][JP][JP][JP][JP][JP]E[JP][JP]E[JP][JP]E
[JP][JP][JP][JP][JP][JP][JP][JP][JP][JP][JP]
[JP][JP][JP][JP][JP][JP][JP][JP][JP][JP][JP]...
[continues for 4096 tokens]
```

Each `[JP]` represents a Japanese hiragana token (e.g., no, wo, kara). This cross-lingual collapse—where the model exits the English token subspace entirely—indicates severe embedding-level corruption beyond simple repetition. (Source: `llama_gptq3bit_gsm8k_100q/checkpoint_00100.jsonl`, idx=99.)

DeepSeek-R1-7B: Self-doubt loop. DeepSeek under W3A16 produces extended reasoning traces where coherent calculation gives way to pathological self-correction. The following excerpt is from idx=50 (Lloyd's eggs problem, pred=252, gt=294):

```
...So, if he sells 252 eggs per day, that's 252
times 252 eggs per day. Wait, no, wait, no, no,
no, no, no, no, no, no, no, no, no, no, no, no,
no, no, no, no, no, no, no, no, no, no, no, no,
no, no, no, no, no, no, no, no, no, no, no, no,
no, no, no, no, no, no, no, no, no, no, no, no...
[continues for 4096 tokens]
```

(Source: `deepseek_w3a16/checkpoint_00100.jsonl`, idx=50.) The model reaches a correct intermediate result (252/12 = 21 dozen $\rightarrow$ \$42/day) but collapses into a "no" loop at the next calculation step, consuming the remaining token budget.

A.13 Answer Extraction Pipeline

GSM8K and MATH-500 answers are extracted from model outputs via a multi-stage fallback pipeline applied in strict priority order:

1. **Boxed format**: Extract the last `\boxed{...}` expression (standard MATH-500 output format).
2. **Hash delimiter**: Extract the numeric value following the last `####` marker (standard GSM8K format).
3. **Natural language**: Match patterns such as "the answer is X", "the final answer is X", "therefore, X" (case-insensitive).

Table 15: Null prediction rates before and after extraction pipeline fix. Post-fix nulls are exclusively from truncated responses.

Benchmark	Model–Precision	Pre-fix %	Post-fix %
GSM8K	DeepSeek FP16	0.0	0.0
GSM8K	14B GPTQ-3bit	18.1	0.0
MATH-500	DeepSeek FP16	7.4	0.0
MATH-500	DeepSeek GPTQ-3bit	48.2	47.6†
MATH-500	Qwen3 FP16	15.8	0.0
HumanEval	DeepSeek FP16	0.0	0.0
HumanEval	DeepSeek AWQ	0.0	0.0
HumanEval	DeepSeek GPTQ-3bit	0.0	0.0

†Remaining nulls are from responses hitting the 8,192-token limit with no answer marker—genuine truncation failures, not extraction errors.

4. **Last numeric fallback**: Extract the last standalone number in the response as a final resort.

Each extraction records which pattern matched, enabling auditing of the pipeline's behavior. If no pattern matches (typically because the response was truncated before producing any answer), the prediction is recorded as `None`.

Null prediction audit. Table 15 reports the null prediction rate (fraction of responses yielding no extractable answer) before and after the pipeline fix. Pre-fix, a simpler extractor (covering only `\boxed{}` and `####`) produced 7–48% null rates on MATH-500 due to models using varied answer formats. Post-fix, remaining nulls correspond exclusively to truncated responses ($\geq$8,000 tokens) that genuinely contain no answer.

A.14 Perplexity Evaluation Configuration

WikiText-2 perplexity (Table 25) is computed with the following configuration:

- **Dataset**: WikiText-2 test split (245 documents, $\sim$250K tokens).
- **Method**: Sliding window with stride 512 tokens, following the standard protocol of Frantar et al. [2023].
- **Sequence length**: Model's maximum context length (up to 4096 tokens per window).
- **Stride**: 512 tokens. Each evaluation window overlaps with the previous by (sequence_length $-$ 512) tokens; only the final 512 tokens of each window contribute to the loss.
- **Loss computation**: Cross-entropy loss averaged over all non-overlapping token positions. Perplexity $= \exp(\text{mean loss})$.
- **Precision**: All activation computations in FP16 (WxA16 regime), matching the inference-time precision. Quantized models are loaded via `GPTQModel` (for GPTQ) or `AutoAWQ` (for AWQ) with dequantization to FP16 activations.
- **Batch size**: 1 (sequential processing to avoid padding artifacts).

A.15 Qwen2.5-14B MATH-500 Difficulty Breakdown

To test whether the difficulty-dependent cliff (Table 7) generalizes beyond DeepSeek, we stratify Qwen2.5-14B's MATH-500 results by difficulty level. Table 16 reports accuracy across the n=370 questions common to FP16, AWQ, and GPTQ evaluations (GPTQ evaluation incomplete; 130 problems excluded).

The difficulty gradient for Qwen2.5-14B is *inconclusive*. While the easy–hard contrast is directionally consistent with DeepSeek's pattern (Level 4–5 average $\Delta_{\text{cliff}}{\approx}23$ pp vs. Level 1–2 average ${\approx}8.5$ pp), two caveats prevent a generalization claim. First, the per-level gradient is non-monotonic: Level 2 ($\Delta = -7.0$ pp) is smaller than Level 1 (-10.0 pp), and Level 5 (-22.5 pp) is smaller

Table 16: Qwen2.5-14B accuracy (%) on MATH-500 by difficulty level (n=370 common questions; GPTQ evaluation incomplete). Δ_{cliff} is the FP16→W3A16 drop. The easy–hard contrast is directionally consistent with DeepSeek (Table 7) but the per-level gradient is non-monotonic; see text for caveats.

Level	n	FP16	W4A16 AWQ	W3A16 GPTQ	Δ_{cliff}
1	30	90.0	86.7	80.0	−10.0
2	71	84.5	88.7	77.5	−7.0
3	78	83.3	76.9	69.2	−14.1
4	102	71.6	61.8	48.0	−23.5
5	89	50.6	40.4	28.1	−22.5
All	370	73.0	67.0	55.9	−17.0

Table 17: GSM8K accuracy under GPTQ W4A16 vs. AWQ W4A16 (`float16` dtype). AWQ–GPTQ differences at W4A16 are $\leq$ 1.4 pp across both models, confirming the W4→W3 cliff is precision-driven, not method-driven.

Model	Method	Precision	n	Acc (%)	Δ vs. FP16 (pp)
DeepSeek-R1-7B	FP16 (baseline)	16-bit	450	78.0	—
	AWQ	W4A16	450	75.4	−2.6
	GPTQ	W4A16	500	74.6	−3.4
	GPTQ	W3A16	450	54.2	−23.8
Qwen3-8B	FP16 (baseline)	16-bit	500	93.6	—
	AWQ	W4A16	500	92.4	−1.2
	GPTQ	W4A16	500	92.2	−1.4
	GPTQ	W3A16	500	0.0	−93.6

DeepSeek AWQ–GPTQ difference at W4A16: 0.8 pp; Qwen3: 0.2 pp. Both non-significant, confirming W4A16 near-losslessness is method-independent. The W4→W3 cliff (−20.4 to −93.6 pp) dwarfs any method difference. Qwen3 GPTQ W4A16 was evaluated with `max_new_tokens`=2048 (vs. 4096 for other GSM8K runs); truncation rate was 0%, so results are unaffected.

than Level 4 (−23.5 pp). Second, the GPTQ evaluation is incomplete (n=370/500), reducing per-level sample sizes to as few as n=30 (Level 1) and introducing potential selection bias from the 130 excluded problems. We report these results for transparency but do not claim that difficulty-dependent amplification generalizes to Qwen2.5-14B; a complete evaluation with larger per-level samples would be needed to draw that conclusion. W4A16 AWQ also shows degradation at the hardest levels (61.8% and 40.4% at Levels 4–5 vs. FP16's 71.6% and 50.6%), consistent with the Qwen2.5-14B MATH-500 W4A16 exception noted in §4.1.

A.16 GPTQ W4A16 Control Experiment

To rule out the possibility that the W4(AWQ)→W3(GPTQ) cliff is an artifact of switching quantization methods, we evaluate DeepSeek-R1-Distill-Qwen-7B and Qwen3-8B under GPTQ W4A16 with matched settings (group size 128, n=500).

Dtype artifact. Initial GPTQ W4A16 evaluation used `bfloat16` compute dtype, yielding 83.4% accuracy (n=500, truncation 7.2%)—apparently 8.0 pp above AWQ W4A16 (75.4%). Switching to matched `float16` dtype (consistent with AWQ's inference path) reduced accuracy to 74.6% (truncation 26.6%), closing the gap to 0.8 pp. The 8.8 pp `bfloat16`→`float16` difference is a dtype artifact: `bfloat16`'s wider dynamic range partially compensates for quantization error, inflating accuracy. All GPTQ W4A16 results in the paper use the `float16`-matched evaluation.

Profile-shift cross-validation. We additionally compare the 4D functional profile (sink concentration, entropy, Gini, positional bias) between GPTQ W4A16 and AWQ W4A16 on the same DeepSeek model. The per-dimension profile differences between the two W4A16 methods are 21–34% of the W3A16→FP16 shift magnitude (sink: 21.5%, entropy: 21.0%, Gini: 33.8%, positional

Table 18: McNemar contingency tables for FP16 vs. W3A16 on GSM8K. a: both correct; b: FP16 correct, W3A16 incorrect (flip); c: FP16 incorrect, W3A16 correct (recovery); d: both incorrect. McNemar's test is computed on the discordant pair counts (b, c).

Model	n	a	b	c	d	χ^2	p
DeepSeek-R1-7B	450	243	108	1	98	105.0	$<10^{-20}$
Qwen2.5-14B	1319	850	384	13	72	346.7	$<10^{-20}$
Phi-3-medium-14B	500	324	133	2	41	127.0	$<10^{-20}$
Qwen2.5-7B	1319	1011	188	31	89	112.6	$<10^{-20}$
Gemma-2-9B	500	384	34	15	67	7.4	0.007

All five tests are significant at $\alpha_{\text{adj}} = 0.01$ (Bonferroni-corrected for 5 tests). Across all models, correct→incorrect flips (b) vastly exceed recoveries (c), with b/c ratios ranging from $2.3\times$ (Gemma-2) to $108\times$ (DeepSeek). DeepSeek uses the 450-question matched subset (shared across FP16/W4A16/W3A16); both Qwen2.5 models use the full GSM8K test set (n=1,319); remaining models use n=500 from the first 500 GSM8K test questions. Minor discrepancies with full-set accuracy in Table 1 reflect subset sampling: Phi-3 is 65.2% here vs. 64.8% full-set, and Gemma-2 is 79.8% here vs. 80.2% full-set (both ≤ 0.8 pp).

Table 19: McNemar's paired test for W4A16 AWQ vs. FP16 across three models and three benchmarks (k=9 tests, Bonferroni $\alpha_{\text{adj}} = 0.0056$). No test survives Bonferroni correction.

Model	Benchmark	n	b	c	Δ pp	p	p_{adj}	Sig.
DeepSeek	GSM8K	499	62	50	-0.6	.299	1.00	n.s.
Qwen3	GSM8K	500	15	9	-1.2	.307	1.00	n.s.
Llama	GSM8K	500	31	24	-1.4	.419	1.00	n.s.
DeepSeek	MATH-500	500	34	42	$+1.6$	.422	1.00	n.s.
Qwen3	MATH-500	500	19	20	$+0.2$	1.00	1.00	n.s.
Llama	MATH-500	500	61	38	-4.6	.027	.243	n.s.
DeepSeek	HumanEval	164	15	7	-4.9	.136	1.00	n.s.
Qwen3	HumanEval	164	12	20	$+4.8$	.216	1.00	n.s.
Llama	HumanEval	164	24	12	-7.3	.067	.603	n.s.

$p_{\text{adj}} = \min(p \times 9,\, 1.0)$. The sole uncorrected-significant result (Llama MATH-500, $p = .027$) does not survive Bonferroni correction ($p_{\text{adj}} = .243$). Llama's consistent negative trend across all three benchmarks (-1.4, -4.6, -7.3 pp) may warrant investigation at larger sample sizes.

bias: 30.0%), confirming that the quantization-method confound is small relative to the bit-width effect that drives the cliff.

A.17 FP16 vs. W3A16 McNemar Contingency Tables

Table 18 reports the full 2×2 McNemar contingency tables for the five non-collapsed models on GSM8K under FP16 vs. W3A16 GPTQ. All five models show highly significant asymmetry ($p < 0.001$), with correct→incorrect flips vastly exceeding incorrect→correct recoveries.

A.18 W4A16 McNemar Tests: Full Statistical Detail

Table 19 reproduces and extends the W4A16 vs. FP16 McNemar tests from the main text with explicit Bonferroni-adjusted significance thresholds. Nine simultaneous tests yield $\alpha_{\text{adj}} = 0.05/9 = 0.0056$.

A.19 Perturbation Replay: Non-Truncated Accuracy

Table 20 reports accuracy restricted to non-truncated responses for the DeepSeek perturbation replay experiment, isolating direct reasoning errors from truncation-induced failures.

Table 20: DeepSeek-R1-7B perturbation replay: accuracy on non-truncated responses only (those not hitting the 4096-token limit). Once truncation-induced failures are excluded, perturbation has minimal impact on reasoning accuracy—the damage is almost entirely mediated by the truncation cascade.

Condition	n	Trunc (%)	Non-trunc acc (%)	Full acc (%)
FP16 baseline	500	2.6	91.4	90.0
attention_only	500	4.4	91.2	88.0
mlp_only	500	13.6	93.1	83.4
all_weights	500	30.0	90.6	70.8

Non-truncated accuracy ranges from 90.6% to 93.1% across all conditions (vs. 91.4% baseline), indicating that when the model produces a complete answer under perturbation replay, the answer is almost always correct. The accuracy drop under `all_weights` (-19.2 pp) is therefore attributable to the truncation cascade: the replayed quantization error destabilizes the generation length distribution, causing more responses to exhaust the token budget before converging on an answer.

Table 21: ARC-Challenge accuracy (%) under FP16 and W3A16 GPTQ (n=1172, 25-shot, log-likelihood scoring). Four of five models exceed 85% FP16 accuracy, compressing the dynamic range for most models. The exception is DeepSeek-R1-7B (FP16 $= 72.0\%$), which shows the largest Δ and is the only model where ARC damage exceeds MMLU damage.

Model	FP16	W3A16 GPTQ	Δ (pp)
Qwen2.5-14B	94.3	91.0	-3.2
Phi-3-medium-14B	92.6	85.6	-7.0
DeepSeek-R1-7B	72.0	58.2	-13.8
Qwen2.5-7B	90.4	85.2	-5.3
Gemma-2-9B	90.0	86.4	-3.6

For the four models with FP16 $> 85\%$, ARC Δ values (-3.2 to -7.0 pp) are compressed relative to GSM8K cliff magnitudes, consistent with a ceiling effect that limits observable degradation. DeepSeek-R1-7B, the one model below the ceiling (FP16 $= 72.0\%$), shows a substantial -13.8 pp drop—the ARC RE (ARC Δ/MMLU $\Delta = 13.8/9.5 = 1.45$) exceeds 1.0, consistent with its cliff-regime classification. We treat ARC as confirmatory rather than primary given the ceiling compression affecting most models.

A.20 ARC-Challenge Results

Table 21 reports ARC-Challenge 25-shot accuracy (log-likelihood scoring) for the five non-collapsed models under FP16 and W3A16 GPTQ (n=1172).

A.21 Perturbation Replay: Full Three-Model Comparison

Table 22 consolidates all perturbation replay conditions across the three tested models at scale c=1.0.

A.22 Statistical Methods

We use the following procedures throughout, with all tests two-sided at $\alpha = 0.05$ unless noted.

McNemar's test. For paired binary correctness between two precision levels (e.g., FP16 vs. W4A16 or FP16 vs. W3A16) on the same question set, we use McNemar's exact test on the 2×2 contingency of discordant pairs. This is the appropriate test for paired nominal data and avoids the violated independence assumption of an unpaired χ^2 test.

Bootstrap confidence intervals. Accuracy differences (Δ) and the Reasoning Excess ratio RE are reported with 95% confidence intervals from 10,000 bootstrap resamples of the paired per-question correctness data.

TOST (Two One-Sided Tests) equivalence testing. For conditions hypothesized to be *equivalent* to baseline (e.g., mlp_only perturbation equals FP16), a non-significant t-test is insufficient — ab-

Table 22: Perturbation replay results across three models (GSM8K, scale c=1.0). DeepSeek and Gemma-2 use the full four-condition matrix plus Gaussian baseline; Phi-3 uses the updated evaluation (n=500, 0-shot) after fixing a dequant implementation bug.

Model	Condition	n	Acc (%)	Δ (pp)	Trunc (%)
DeepSeek-R1-7B	FP16 baseline	500	90.0	—	2.6
	attention_only	500	88.0	−2.0	4.4
	mlp_only	500	83.4	−6.6	13.6
	all_weights	500	70.8	−19.2	30.0
	Gaussian‡	200	19.0	−71.0	93.0
Gemma-2-9B	FP16 baseline	500	83.6	—	0.0
	attention_only	200	80.0	−3.6	0.0
	mlp_only	200	79.5	−4.1	0.0
	all_weights	500	78.4	−5.2	0.0
Phi-3-medium-14B	FP16 baseline	500	84.2	—	0.0
	all_weights	500	57.4	−26.8	1.2

Reference GPTQ cliffs: DeepSeek −23.8 pp, Gemma-2 −3.4 pp, Phi-3 −26.6 pp. Phi-3 uses 0-shot evaluation (8-shot FP16 baseline is 91.4%; Table 1). ‡Gaussian: i.i.d. $\mathcal{N}(0, \sigma_\ell^2)$ with $\sigma_\ell = \mathrm{std}(\Delta W^{(\ell)})$; bias perturbation disabled. **Note:** This table uses the original evaluation protocol (n=500 baseline, n=200 Gaussian, default seed) for DeepSeek/Gemma-2. Matched-setting results (seed=42/123, n=500) in Table 3 show Gemma-2 Gaussian/ΔW ratio = 3.3–5.6× (three seeds).

sence of evidence is not evidence of absence. We use TOST with an equivalence bound of $\Delta = 5$ percentage points: two one-sided tests at $\alpha = 0.05$ jointly reject the null of non-equivalence if the 90% CI of the accuracy difference lies fully within $[-5, +5]$ pp.

Bonferroni correction. For families of simultaneous tests—the 9 model×benchmark W4A16 McNemar comparisons in C1 (3 models × 3 benchmarks) and the 5 chain-length×flip-rate Spearman correlations in C3 (one per non-collapsed model)—we apply Bonferroni correction to the significance threshold ($\alpha_{\text{adj}} = 0.05/k$). Both uncorrected and corrected p-values are reported where applicable.

Spearman rank correlation. For non-parametric associations (e.g., profile shift magnitude vs. Fisher information, chain length vs. flip rate), we report Spearman's ρ, which is robust to the non-Gaussian, long-tailed distributions observed in per-head sensitivity measures.

Effect sizes. Alongside p-values we report effect sizes in interpretable units: accuracy differences in percentage points (pp), ratios for Reasoning Excess, and Cohen's h for proportion differences in flip-rate analyses. This ensures that statistical significance is not conflated with practical significance under large n.

A.23 Methodology: Scope and Verification Details

Scope caveat. Our W3A16 results characterize *GPTQ-specific* behavior, not the fundamental 3-bit precision limit. Methods such as QuIP# [Tseng et al., 2024] and AQLM [Egiazarian et al., 2024] may recover more accuracy at equivalent bit-width through incoherence processing and vector quantization respectively. We therefore interpret our findings as characterizing the precision threshold of "the most widely deployed 3-bit method", which remains the practically relevant regime for near-term deployment.

Collapse verification via dequantization. For models exhibiting complete reasoning collapse under W3A16 (Llama-3.1-8B and Qwen3-8B), we perform a dequantization control: we dequantize the GPTQ 3-bit weights back to FP16 storage format ($W_{\text{dequant}} = s \cdot q + z$ for each group) and re-evaluate on GSM8K. If collapse persisted under the dequantized FP16 weights, it must stem from the quantization error itself rather than any kernel, attention, or toolchain artifact specific to the 3-bit runtime. This rules out trivial explanations and localizes the damage to the weight perturbation $\Delta W = W_{\text{dequant(W3)}} - W_{\text{FP16}}$. We note that dequantization verification was performed only for the

Table 23: MATH-500 accuracy (%) across five models and precision tiers (n=500 per condition except where noted). Δ_{cliff} is the FP16→W3A16 drop.

Model	FP16	W4A16 AWQ	W3A16 GPTQ	Δ_{cliff}
Llama-3.1-8B	43.4	38.8	0.0*	−43.4
Qwen3-8B	78.6	78.8	0.0†	−78.6
DeepSeek-R1-7B	76.4	78.0	41.6	−34.8
Gemma-2-9B	46.6	—	35.4	−11.2
Qwen2.5-7B	69.4	—	43.6	−25.8

*Early-stopped at n=40: 0% accuracy, 100% truncation. †Early-stopped at n=4: 0% accuracy, 100% truncation.
W3A16 truncation rates (generation ≥ `max_new_tokens`): DeepSeek 51.2%, Qwen2.5-7B 5.8%, Gemma-2 0%, FP16 baselines $\sim 0\%$. Truncated outputs are scored normally (not excluded); actual accuracy may be slightly underestimated for high-truncation models.

Table 24: HumanEval pass@1 (%) across five models and precision tiers (n=164). Δ_{cliff} is the FP16→W3A16 drop.

Model	FP16	W4A16 AWQ	W3A16 GPTQ	Δ_{cliff}
Llama-3.1-8B	53.0	45.7	0.0*	−53.0
Qwen3-8B	72.0	76.8	0.0†	−72.0
DeepSeek-R1-7B	79.3	74.4	27.4	−51.9
Phi-3-medium-14B	57.3	—	19.5	−37.8
Gemma-2-9B	61.6	61.0	42.1	−19.5

*Early-stopped at n=41: 0% pass@1, 100% truncation. †Early-stopped at n=1: 0% pass@1, output is degenerate repetition.

two collapse models; whether the partial degradation in cliff-regime models (DeepSeek, Qwen2.5-14B, Phi-3) is similarly free of toolchain artifacts remains unverified, though the GPTQ W4A16 control (Appendix A.16) provides indirect evidence that GPTQ's inference path does not introduce systematic bias.

A.24 MATH-500 and HumanEval Results

MATH-500. We validate the cliff on MATH-500 (500 competition problems, `max_new_tokens` = 8192; Table 23). The cliff replicates: Llama and Qwen3 collapse to 0% under W3A16, and DeepSeek drops from 76.4% (FP16) to 41.6% ($\Delta_{\text{cliff}} = -34.8$ pp)—a steeper cliff than GSM8K's -23.8 pp, consistent with competition problems eliciting longer reasoning chains that accumulate more quantization error. Gemma-2's MATH-500 cliff (-11.2 pp) is $3.3\times$ its GSM8K cliff (-3.4 pp), demonstrating that "mild" is task-dependent: the same architecture that resists degradation on grade-school problems shows substantial damage on competition-level problems requiring longer chains. Qwen2.5-7B—the mildest GSM8K cliff (-11.9 pp)—drops 25.8 pp on MATH-500,[4] with a clear difficulty gradient: Level 1 accuracy degrades from 93.0% to 72.1% while Level 5 falls from 45.5% to 17.9%, consistent with longer, harder chains accumulating more error. Truncation on DeepSeek W3A16 rises from 13.4% (W4A16) to 51.2%, mirroring the output-bloat pattern observed on GSM8K.

HumanEval. To test cross-domain generality, we evaluate five models on HumanEval [Chen et al., 2021] (n=164; Table 24). The W3A16 cliff extends to code generation: Llama and Qwen3 collapse to 0% (pass@1), DeepSeek drops 51.9 pp, Phi-3 drops 37.8 pp, and Gemma-2 drops 19.5 pp ($5.7\times$ its GSM8K cliff). Gemma-2's HumanEval result is particularly informative: despite being "mild" on GSM8K ($\Delta = -3.4$ pp), it loses nearly a third of its code-generation capability under W3A16, further confirming that mild-regime status is task-dependent. W4A16 differences are not individually significant at n=164, though Llama's consistent negative trend (-7.3 pp, $p = 0.067$) extends its pattern across benchmarks.

[4] MATH-500 W3A16 evaluation had a 5.8% truncation rate (29/500 responses exceeded the generation limit). Non-truncated accuracy may differ slightly.

Table 25: WikiText-2 perplexity (sliding window, stride 512) across three precision tiers. Perplexity does not reliably indicate reasoning collapse.

Model	FP16		W4A16 AWQ		W3A16 GPTQ	
	PPL	$\Delta\%$	PPL	$\Delta\%$	PPL	$\Delta\%$
Llama-3.1-8B	6.43	—	6.75	+5.1	328.08	+5003
Qwen3-8B	8.57	—	8.90	+3.8	NaN[†]	—
DeepSeek-R1-7B	20.73	—	22.53	+8.7	27.74	+33.8

[†]Qwen3-8B W3A16 GPTQ produces NaN loss on WikiText-2: 3-bit quantization yields numerically unstable logits during the teacher-forced forward pass, causing overflow in the cross-entropy computation.

HumanEval RE. HumanEval provides a second RE axis beyond mathematics. DeepSeek-R1-7B's HumanEval RE reaches 5.46 ($\Delta_{\text{HumanEval}} = -51.8$ pp vs. $\Delta_{\text{MMLU}} = -9.5$ pp), more than double its GSM8K RE of 2.51. Gemma-2-9B yields a HumanEval RE of 3.75 ($\Delta_{\text{HumanEval}} = -19.5$ pp vs. $\Delta_{\text{MMLU}} = -5.2$ pp), confirming that code generation is consistently more vulnerable than knowledge recall under W3A16. The remaining models cannot contribute a multi-model RE estimate: Llama-3.1-8B ($\Delta_{\text{HumanEval}} = -53.0$ pp) and Qwen3-8B ($\Delta_{\text{HumanEval}} = -72.0$ pp) both collapse to 0% pass@1, precluding RE computation. With two independent data points (DeepSeek $5.46\times$, Gemma-2 $3.75\times$), the pattern is consistent: generative tasks requiring multi-step coherence—whether mathematical or code generation—are disproportionately damaged by aggressive quantization.

A.25 Perplexity vs. Reasoning Accuracy

Perplexity is the standard quantization-quality metric [Frantar et al., 2023, Lin et al., 2024], yet it fails to reliably indicate reasoning collapse (Table 25). W4A16 raises WikiText-2 perplexity by 3.8–8.7% relative to FP16—consistent with near-lossless reasoning. Under W3A16 the picture diverges: DeepSeek-R1-7B's perplexity rises by only 33.8% (20.73→27.74) despite losing 23.8 pp on GSM8K; Llama-3.1-8B's perplexity explodes (+5003%), consistent with complete collapse; Qwen3-8B produces NaN from numerical overflow. The DeepSeek case is decisive: a moderate perplexity increase masks a severe reasoning deficit, confirming that perplexity changes are poorly calibrated to reasoning damage—a single-token next-word metric is an unreliable proxy for multi-step coherence.

A.26 Profile Shift Under Quantization

To link the accuracy cliff to measurable changes in attention-head behavior, we compute the 4D functional profile (sink concentration, entropy, Gini, positional bias; §3.6) and report the mean per-dimension $|z|$-scored shift $\overline{|z|}$ for DeepSeek-R1-7B across pairwise transitions (Table 26). W4A16 produces near-zero shift ($\overline{|z|} \leq 0.31$, no head exceeds $|z|>2$). W3A16 shifts are 3–4$\times$ larger across every dimension: entropy rises to $\overline{|z|} = 1.04$ ($3.9\times$), indicating systematically broadened attention. The damage concentrates in the W4→W3 step: $\overline{|z|}_{\text{entropy}}$ for W4A16→W3A16 (0.84) is of the same order as the full FP16→W3A16 shift (1.04), confirming that W4A16 acts as a near-transparent stage and the cliff arises from the final bit removal. The shift is distributed across functional dimensions (12 heads reach $|z|>2$ in FP16→W3A16; entropy-dominant heads account for 11 of these), not concentrated on one axis.

Profile-shift magnitude and Fisher information show weak, non-significant correlation (Spearman $\rho = 0.14$, $p = 0.15$, N=112 KV head groups; Table 27), suggesting that Fisher-guided mixed-precision methods [Guan et al., 2024] may not reliably capture the functional disruption associated with the cliff.

A.27 Perturbation Replay: Module Decomposition and Head-Level Analysis

Decomposition: attention vs. MLP. Within each model, the attention-only and MLP-only conditions reveal where perturbation damage accumulates. For DeepSeek, MLP-only damage (-6.6 pp, truncation 13.6%) is $3.3\times$ larger than attention-only damage (-2.0 pp, truncation 4.4%); perturb-

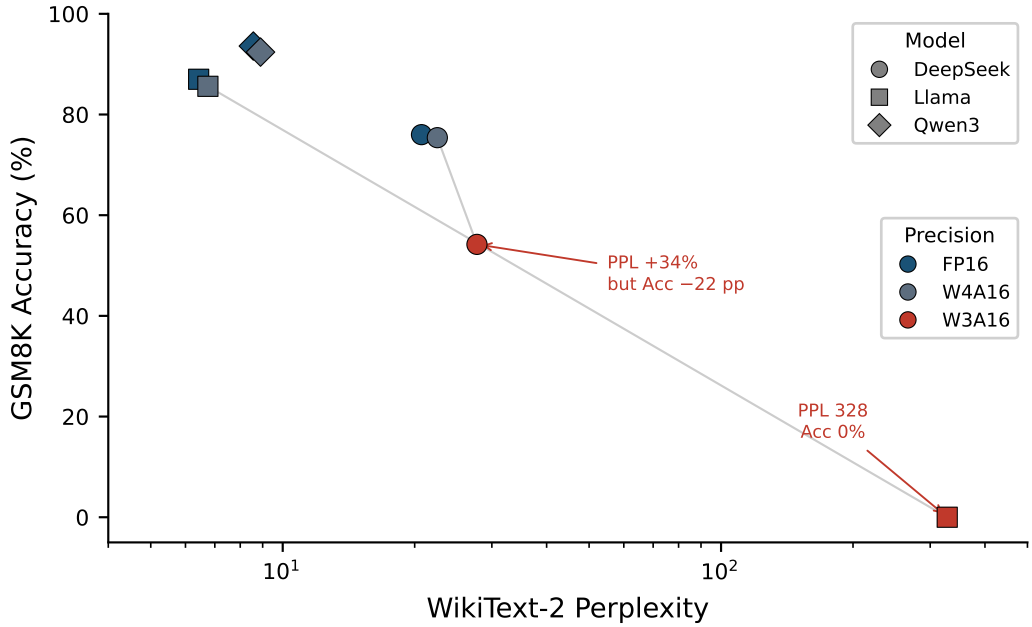

Figure 4: Perplexity vs. reasoning accuracy (log x-axis). Perplexity changes are disproportionate to accuracy loss: DeepSeek W3A16 shows $+33.8\%$ PPL despite losing 23.8 pp on GSM8K, while Llama W3A16 explodes to PPL $= 328$. Qwen3 W3A16 is omitted (PPL diverges to NaN).

Table 26: Mean per-dimension $|z|$-scored profile shift for DeepSeek-R1-7B. W4A16 produces negligible shift; W3A16 shifts are 3–4× larger and drive 12 heads above $|z|>2$.

Transition	Sink $\overline{|z|}$	Entropy $\overline{|z|}$	Gini $\overline{|z|}$	Bias $\overline{|z|}$	Sig. heads
FP16 → W4A16	0.31	0.27	0.15	0.14	0
FP16 → W3A16	0.90	1.04	0.58	0.57	12
W4A16 → W3A16	0.63	0.84	0.52	0.47	7

Significant heads: $|z|>2$ in any dimension. Entropy-dominant heads account for 11 of 12 significant heads in the FP16→W3A16 comparison.

ing MLP weights triggers the truncation cascade that attention-only perturbation does not. For Gemma-2, attention-only and MLP-only are comparable (-3.6 pp vs. -4.1 pp), and neither triggers any truncation. Neither partial injection reproduces the full `all_weights` damage, indicating that per-module perturbation adds rather than saturates—consistent with the distributed-damage picture from the profile-shift analysis (Appendix A.26).

Distributed damage (head-level replication). We additionally verify distribution at the attention-head level on DeepSeek: replaying W3 error into the five KV heads with largest FP16→W3A16 profile shift (`target`) leaves GSM8K accuracy at 89.6%; doing the same for five minimal-shift control heads leaves it at 90.0%; target+control leaves it at 89.8%; all-heads simultaneously drops to 87.2% ($p = 0.040$). Two one-sided tests (TOST, ± 5 pp bound) establish formal equivalence of all three partial-injection conditions to baseline ($p < 0.01$ each). Neither maximally- nor minimally-shifted heads carry a disproportionate share of the damage—the cliff emerges only when all head groups are perturbed simultaneously. This head-level distribution mirrors the module-level distribution above: mitigation strategies that protect a small fraction of parameters (heads or modules) cannot prevent the cliff, because the unprotected bulk contributes enough perturbation on its own.

Confound control: generation length. Attention entropy correlates strongly with generation length ($r = 0.939$) across GSM8K samples. After controlling for generation length via partial correlation, entropy's apparent predictive power over per-sample accuracy vanishes (partial $r = -0.043$,

Table 27: Spearman rank correlation between functional profile shift magnitude and Fisher information for DeepSeek-R1-7B (112 KV head groups).

Model	Heads	Fisher Mean	Fisher Max	ρ	p-value
DeepSeek-R1-7B	112	7.20×10^{-6}	4.11×10^{-4}	0.14	0.15

n.s.), while late-layer attention Gini retains independent validity (partial $r = 0.185$, $p < 0.001$; Bonferroni-adjusted $\alpha_{\text{adj}} = 0.0125$). The near-complete truncation rate (99–100%) in Llama and Qwen3 under W3A16 is the extreme of this coupling: disrupted attention drives indefinite generation, which mechanically elevates entropy.

A.28 Dose-Response Details

Sweeping $c \in \{0.25, 0.5, 1.0, 2.0\}$ yields a dose-response curve relating perturbation magnitude to accuracy degradation (Figure 2).[5] Gemma-2 degrades monotonically: from 82.0% at c=0.25 through 80.5% at c=0.5 to 78.4% at c=1.0 (n=200 for sub-1.0 scales; n=500 for c=1.0; the n=200 subset FP16 baseline is 82.0%), a total drop of only 5.2 pp over a $4\times$ magnitude increase. DeepSeek shows an overall declining trend but exhibits a striking non-monotonicity: accuracy drops to 88.0% at c=0.25 (n=200), then *rises* to 91.0% at c=0.5 (n=300, seed 42)—above the n=200 subset FP16 baseline (89.0%, $+2.0$ pp)—before collapsing to 70.8% at c=1.0 (n=500). The injection is deterministic (scaled ΔW with greedy decoding) and the non-monotonicity is reproducible, indicating that the effect of perturbation along the quantization-error direction is not a simple function of magnitude. At c=2.0 (twice the real GPTQ error magnitude), both models collapse: DeepSeek drops to 0.0% (n=200, 87.5% truncation) and Gemma-2 to 0.5% (n=190, 0% truncation), confirming that sufficient perturbation magnitude eventually overwhelms even mild-regime architectures—though through qualitatively different failure modes (§5.3). The steep DeepSeek curve between c=0.5 and c=1.0 is consistent with a threshold phenomenon: as perturbation magnitude approaches the real GPTQ level, the truncation cascade activates (truncation rises from 2.6% to 30.0%), sharply amplifying accuracy loss. Gemma-2's monotonic curve, by contrast, reflects direct reasoning damage without any truncation involvement.

A.29 Phi-3 Perturbation Replay

Phi-3-medium-14B structured replay uses a 0-shot evaluation protocol (FP16 baseline 84.2%, n=500). Under `all_weights` perturbation at c=1.0, accuracy drops to 57.4% (-26.8 pp), closely matching the actual GPTQ W3A16 cliff (-26.6 pp, within 0.2 pp) with only 1.2% truncation. An earlier attempt terminated at n=50 due to a dequant implementation bug (`g_idx` handling for `desc_act=true`); the results reported here are from the corrected re-run. This places Phi-3 in the "close match" category alongside Gemma-2, distinct from DeepSeek's generation collapse under Gaussian noise.

A.30 DeepSeek Gaussian Variance Analysis

DeepSeek-R1-7B exhibits substantially higher cross-seed variance under Gaussian perturbation replay than Gemma-2 or Phi-3. Four seeds at c=1.0 yield accuracies of 26.2%, 11.2%, 5.0%, and 19.8% (SD $=$ 9.3 pp; 95% CI $[0.7\%, 30.4\%]$, t-distribution, df=3), compared to Gemma-2's SD $=$ 3.9 pp across three seeds.

Two factors likely explain this sensitivity. First, DeepSeek's long reasoning chains (3,755 $\pm$ 1,022 tokens at FP16) provide more opportunity for stochastic noise to accumulate and interact with the model's self-correction mechanisms; different noise realizations trigger the truncation cascade at different points in the reasoning chain, producing large outcome variance. Second, at 4096-token generation budget, 91–99% of Gaussian-perturbed responses hit the token limit. Near-ceiling truncation creates a regime where small differences in noise realization—affecting whether the model escapes a self-doubt loop before the budget runs out—produce large accuracy swings. By contrast,

[5] Dose-response data were collected under an earlier evaluation protocol (n=200, default seed) that preceded the matched-setting evaluations in Table 3 (n=500, seed=42), except DeepSeek c=0.5 which was re-evaluated at n=300, seed=42. Numerical values are not directly comparable to matched-setting results; see Limitation (ix).

Table 28: Error accumulation model fit across five non-collapsed models. The model fits well for three models with significant chain-length correlation (binned $R^2 = 0.65$–0.87) but fails for the two saturated models.

Model	ε ($\times 10^{-3}$)	95% CI ($\times 10^{-3}$)	McFadden R^2	Binned R^2	n_{samples}
Qwen2.5-7B	0.60	[0.52, 0.69]	0.034	0.65	808
Gemma-2-9B	0.63	[0.43, 0.82]	0.046	0.66	418
Phi-3-medium-14B[†]	1.74	[1.46, 2.03]	0.029	0.87	632
DeepSeek-R1-7B	0.33	[0.28, 0.39]	-0.001	-0.11	390
Qwen2.5-14B	1.26	[1.13, 1.39]	-0.029	-11.97	649

[†]Phi-3 flip data from GPTQ 8-shot evaluation, not 0-shot replay protocol. n_{samples}: number of FP16-correct questions used for fitting.

Gemma-2's shorter chains and zero truncation rate under Gaussian noise yield stable accuracy estimates across seeds.

Despite this variance, the qualitative conclusion is robust: all four DeepSeek seeds show Gaussian damage far exceeding structured damage (3.2–$4.3\times$ per-seed ratios), and the mean ratio ($3.8\pm0.5\times$) is consistent with the 3–$5\times$ range observed across all three models.

A.31 Error Taxonomy Results

Table 10 (main text, Section A.1) reports the error taxonomy. We note a distinction between two token-limit metrics: 56.9% of W3A16 errors hit the generation ceiling (the model continues generating when truncated), but 0% are flagged as `is_truncated` in the error taxonomy because an answer marker is always present—the model produces an answer, but the answer has been corrupted by preceding loops.

A.32 Generation-Length Confound

Attention entropy correlates strongly with generation length across all models ($r = 0.939$, measured across question-level means). Longer generations mechanically increase entropy by spreading attention over more tokens. This confound means that raw entropy-accuracy correlations partly reflect a length-accuracy relationship rather than a direct attention-quality effect. We therefore do not claim entropy as an independent predictor of quantization damage; length-controlled analyses are required to disentangle the two. For this reason, attention entropy appears in this paper only as a descriptive statistic within the profile-shift analysis (Appendix A.26) and is excluded from the four main contributions, none of which depend on entropy as an explanatory variable.

A.33 Error Accumulation Model

We fit a simple per-token error-accumulation model to the flip data from Section 4.3. For each FP16-correct question, we observe a binary outcome (flip or non-flip under W3A16) and the FP16 generation length n (in tokens). Assuming each token independently introduces an error with probability ε, the probability of at least one error in a chain of length n is $P(\text{flip} \mid n) = 1-(1-\varepsilon)^n$. We fit ε via maximum likelihood (equivalent to logistic regression on the complementary log-log link) and report both individual-level McFadden R^2 and binned R^2 (10 equal-frequency bins).

The low individual-level McFadden R^2 (0.03–0.05 for the three fitting models) reflects the stochastic nature of per-question outcomes: any single question's flip depends on which specific tokens are affected, not just chain length. The binned R^2 captures the systematic trend across chain lengths, confirming that the i.i.d. per-token model is a useful first-order approximation for models in the accumulation regime.

Phi-3 exhibits the highest per-token error rate ($\varepsilon = 1.74\times10^{-3}$), approximately $3\times$ that of Qwen2.5-7B (0.60×10^{-3}), consistent with its higher flip rate (29.1% vs. 15.7%).

The two models that do not fit the accumulation model—DeepSeek-R1-7B and Qwen2.5-14B—exhibit qualitatively different failure patterns. Qwen2.5-14B shows no chain-length correlation

Table 29: Gemma-2-9B layer-group perturbation replay (n=300). Damage concentrates in front and middle layers; back layers are essentially unaffected. The sum of individual group effects (-5.7 pp) exceeds the full-model effect (-3.4 pp), indicating partial inter-layer compensation when all layers are perturbed simultaneously.

Layer group	Layers	Acc (%)	Δ (pp)
Baseline (FP16)	—	83.3	—
Front 1/3	0–13	81.3	−2.0
Middle 1/3	14–27	79.3	−4.0
Back 1/3	28–41	83.7	+0.3
Full (all layers)	0–41	80.6[†]	−3.4[†]

[†]Full-model result from Table 3 (n=500); the n=300 baseline here is 83.3% vs. 84.0% for n=500, a minor subset difference.

whatsoever (Spearman $\rho = -0.03$, $p = 0.34$), suggesting uniform vulnerability regardless of chain length. DeepSeek-R1-7B shows a non-monotonic pattern: flip rate peaks at intermediate chain lengths ($\sim$1000–2000 tokens) and decreases for the longest chains ($>$ 2000 tokens), inconsistent with the monotonically increasing prediction of the i.i.d. model. Both patterns are consistent with damage that has saturated beyond the gradual-accumulation regime, where additional chain length no longer incrementally increases flip probability.

A.34 Layer-Group Sensitivity Analysis

To localize which layers contribute most to the quantization cliff, we partition Gemma-2-9B's 42 transformer layers into three equal groups and inject the structured GPTQ error ΔW into each group independently, evaluating on GSM8K (n=300, seed=42).

Front and middle layers account for the entirety of the cliff (-2.0 and -4.0 pp respectively), while back layers show no degradation (+0.3 pp, within noise). The sum of individual effects (-5.7 pp) exceeds the full-model cliff (-3.4 pp), revealing partial inter-layer compensation: when all layers are perturbed simultaneously, interactions between layer groups partially cancel the damage that each group inflicts in isolation. This non-additivity implies that effective mitigation could target the front and middle layers specifically—e.g., via mixed-precision quantization allocating higher bit-widths to layers 0–27—without needing to protect the back layers.

A.35 Broader Impact

Hidden reasoning degradation. The disconnect between perplexity and reasoning accuracy (Table 25) has concerning implications for deployment. A quantized model may appear healthy by standard quality metrics while its multi-step reasoning has collapsed—DeepSeek's perplexity rises only 33.8% at W3A16 while GSM8K accuracy drops 23.8 pp. In safety-critical domains (medical reasoning, legal analysis, financial decision-making), standard deployment monitoring based on perplexity or single-token metrics would give a false "all clear" signal. We recommend that any deployment of sub-4-bit quantized models in reasoning-critical settings include direct evaluation on multi-step reasoning benchmarks, not perplexity alone.

Reproducibility. Code and evaluation scripts will be released upon acceptance.

A.36 Limitations

We acknowledge eleven caveats. (i) *Limited method coverage*: our cliff spectrum is characterized primarily for GPTQ, with single-model RTN and HQQ comparisons (§5.4); QuIP#, AQLM, and other sub-4-bit schemes may shift the threshold or alter the tier ordering. (ii) *Scale ceiling*: all models are 7–14B parameters; whether the cliff persists, intensifies, or vanishes at $>$ 70B is untested, though Qwen2.5-7B vs. 14B within the same family provides weak evidence of scale independence. (iii) *Spectrum resolution*: the RE spectrum rests on $N = 5$ non-collapsed data points; denser sampling across more architectures is needed to establish whether the spectrum is continuous or clustered. (iv) *Single seed*: we use greedy decoding (deterministic outputs), but model checkpoint selection is a single draw that does not capture pre-training variance. (v) *No causal mechanism*: we characterize *what* degrades—reasoning-specifically, architecture-dependently—but not *why* certain architectures resist the cliff. (vi) *Partial dequantization verification*: collapse-regime models were dequant-verified to rule out inference-stack artifacts; cliff-regime models were not subjected to the same verification. (vii) *MMLU sample size asymmetry*: MMLU evaluation uses variable total sample sizes across models (n=2,850 for 14B models vs. n=14,042 for 7B–9B models), which affects bootstrap CI width for the RE metric; we report CIs throughout. (viii) *Phi-3 perturbation replay evaluation protocol*: Phi-3 replay uses a 0-shot protocol (baseline 84.2%) rather than 8-shot (Table 1 FP16 91.4%); the -26.8 pp drop is computed relative to the matched 0-shot baseline. (ix) *Dose-response subset baselines*: sub-c=1.0 dose-response experiments use n=200 subsets whose FP16 baselines differ slightly from the full n=500 evaluations; we report subset-matched baselines in the text. (x) *Gaussian noise variance*: DeepSeek Gaussian replay exhibits high cross-seed variance (SD $=9.3$ pp across four seeds, 95% CI $[0.7\%, 30.4\%]$), substantially exceeding Gemma-2's (SD $=3.9$ pp, three seeds). This wide CI means the DeepSeek damage ratio ($3.8 \pm 0.5\times$) should be interpreted with caution; the qualitative conclusion (Gaussian $\gg$ structured damage) is robust, but the precise ratio is less stable than for Gemma-2 or Phi-3. See Appendix A.30 for analysis. (xi) *Cross-environment execution*: DeepSeek c=1.0 perturbation replay seeds were executed across two server environments (westd-40883 and an AutoDL RTX-6000 instance) rather than the standardized westc-45186 environment used for dose-response experiments; we verified identical model files and software versions across environments.

A.37 Prompt Templates

GSM8K (8-shot, generative). We use model-specific chat templates wrapping the question text. The 8 few-shot exemplars are drawn from the GSM8K training split in fixed order. Each exemplar consists of a question followed by a chain-of-thought solution ending with "`#### [answer]`".

Three chat templates are used depending on the model family. Gemma-2 uses its native `apply_chat_template()` from the HuggingFace tokenizer.

Chat Templates

```
ChatML (DeepSeek, Qwen2.5, Qwen3):
  <|im_start|>user\n{question}<|im_end|>
  <|im_start|>assistant\n

Llama-3 (Llama-3.1):
  <|begin_of_text|><|start_header_id|>user
  <|end_header_id|>\n\n{question}<|eot_id|>
  <|start_header_id|>assistant<|end_header_id|>\n\n

Phi-3:
  <|user|>\n{question}<|end|>\n<|assistant|>\n
```

MMLU (5-shot, log-likelihood). Each subject uses 5 exemplars from its dev split. The prompt follows the standard format:

MMLU Prompt Format

```
The following are multiple choice questions (with
answers) about {subject}.

{question}
A. {choice_A}   B. {choice_B}
C. {choice_C}   D. {choice_D}
Answer: {A/B/C/D}

[... 4 more exemplars ...]

{test question}
A. {choice_A}   B. {choice_B}
C. {choice_C}   D. {choice_D}
Answer:
```

Scoring compares log-likelihoods of the four answer tokens (A/B/C/D) at the final position.

Table 30: Within-MMLU subject analysis under W3A16 GPTQ. Δ_R: accuracy drop on reasoning-intensive subjects; Δ_K: drop on knowledge-intensive subjects; within-MMLU RE $= \Delta_R/\Delta_K$. Four of six models show RE > 1, confirming reasoning-specific vulnerability within a single log-likelihood evaluation format.

Model	FP16	W3A16	Δ_R (pp)	Δ_K (pp)	RE
Phi-3-medium-14B	78.2	68.4	-16.5	-6.0	2.76×
Qwen2.5-14B	80.4	74.0	-14.3	-5.8	2.48×
Qwen2.5-7B	74.3	64.8	-10.3	-6.0	1.71×
Gemma-2-9B	72.3	67.0	-4.6	-3.2	1.43×
DeepSeek-R1-7B[a]	54.0	44.5	-11.0	-11.2	0.99×
Llama-3.1-8B[b]	68.3	52.4	-12.8	-13.4	0.96×
Mean			-11.6	-7.6	**1.72**×[c]

[a]FP16 baseline near floor (54%), limiting observable differential.
[b]Largest overall drop (-15.9 pp); quantization damage spreads uniformly across subject categories at this severity.
[c]Mean RE is the arithmetic mean of per-model RE values, not the ratio of mean drops ($11.6/7.6 = 1.53\times$). The per-model mean better reflects typical reasoning vulnerability across architectures.

ARC-Challenge (25-shot, log-likelihood). Follows the same format as MMLU with 25 exemplars from the training split and the subject line "science". Scoring is identical (log-likelihood of answer label tokens).

HumanEval (zero-shot, generative). The function signature and docstring are wrapped in the model-specific chat template with the system instruction "Please complete the following Python function. Return ONLY the function body, no explanation." Pass@1 is evaluated via the standard HumanEval execution harness.

A.38 Within-MMLU Subject Analysis

To isolate reasoning-specific damage within a single evaluation format, we partition MMLU's 57 subjects into *reasoning-intensive* (formal_logic, abstract_algebra, high_school_mathematics, college_mathematics, elementary_mathematics, machine_learning, electrical_engineering, computer_science) and *knowledge-intensive* (world_religions, us_history, marketing, nutrition, professional_law, management, human_aging, sociology) categories. Table 30 reports the per-category accuracy drop under W3A16 GPTQ.

Methodological note. We group MMLU subjects into reasoning-heavy (mathematics, logic, CS) and knowledge-heavy (social science, humanities) categories. The lower FP16 baselines of reasoning subjects (mean $\sim$62% vs. $\sim$82%) work *against* our hypothesis by compressing the available drop range, yet reasoning subjects still show larger absolute degradation, strengthening the conclusion.

A.39 Compute Resources

All experiments were conducted on a server with four NVIDIA RTX PRO 6000 GPUs (96 GB VRAM each). GSM8K evaluation of 500 questions under W3A16 with `max_new_tokens`=4096 takes approximately 7.1 hours (25 502 seconds wall-clock time). FP16 and W4A16 evaluations are faster due to shorter average generation lengths. Profile extraction (attention hook logging over 500 questions) adds <10% overhead to inference time. The perturbation replay experiment runs on the same hardware with comparable wall-clock time per condition.

A.40 Future Work

The opposite-direction replay deviations motivate **structured error decomposition** to localize architecture-dependent vulnerability at sub-module granularity—identifying which weight groups contribute most to chain-of-thought degradation. **Scale extension** to 70B+ models will test whether cliff-prone families (Llama, Qwen3) exhibit the cliff at larger scale or whether increased redundancy provides natural protection. Applying our RE/flip-rate framework to **QuIP# and AQLM** will determine whether the three-tier spectrum is GPTQ-specific or a universal feature of sub-4-bit quantization. The flip-rate instability at preserved aggregate accuracy motivates **reasoning-aware quantization objectives** that explicitly penalize chain-of-thought divergence rather than minimizing layer-wise reconstruction error alone. Extending this analysis to **quantization-aware training** (QAT) methods [Liu et al., 2023, Shao et al., 2023], which may preserve reasoning structure more effectively by optimizing through simulated quantization error, is a natural complement to our PTQ-focused study.

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